

ÉCOLE CENTRALE DE NANTES
LABORATOIRE ICONIC — CHU NANTES

Récapitulatif par sujet

Débruitage EEG ART / ICA / ICLabel (Motor-Imagery)

Comparaison CSP+LDA et MSE, sujet par sujet

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Introduction

Nous avons identifié l'article « *Augmenting brain-computer interfaces with ART : An artifact removal transformer for reconstructing multichannel EEG signals* » de Chuang *et al.*, qui présente une nouvelle méthode de débruitage appelée **ART** (*Artifact Removal Transformer*) et montre de bons résultats sur des données d'imagerie motrice (*motor-imagery*). Dans leur étude, les auteurs entraînent ART sur la base *Eyes-Closed/ Eyes-Open Resting-State* (Chuang *et al.*, 2022). Nous avons identifié une limite à cette base pour l'entraînement : elle ne contient pas les artefacts habituels d'une séance EEG, comme les clignements des yeux (*blinks*).

C'est ce qui motive notre choix d'utiliser la base **Schalk *et al.* (2004), Motor Imagery (BCI2000)**, et d'y entraîner ART en prenant comme signaux propres ceux nettoyés par **ICLabel**. L'objectif est de tester si ART peut apprendre à nettoyer automatiquement les signaux aussi bien, voire mieux, qu'ICLabel.

Pour chaque sujet exclu (protocole *leave-one-subject-out*), ce document rapporte : le taux de classification gauche/droite (CSP+LDA) sur les différents signaux débruités disponibles (brut, ICA manuelle, ICLabel automatique, ART, ...); l'erreur quadratique moyenne (MSE) entre le signal nettoyé par ICA et celui nettoyé par ICLabel; ainsi que l'évolution du MSE (train/eval) pendant l'entraînement *leave-one-subject-out* d'ART.

Mots-clés : EEG, BCI, ART, ICA, ICLabel, CSP, MSE, motor-imagery.

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1 Méthodologie

1.1 Taux de classification (CSP+LDA)

Pour mesurer à quel point l'intention motrice (main gauche vs main droite) reste lisible dans un signal après débruitage, on applique une chaîne de classification standard en EEG : filtrage passe-bande dans la bande sensori-motrice (7–30 Hz), extraction d'une courte fenêtre temporelle après le stimulus, puis un filtrage spatial (CSP) qui recherche les directions de l'espace séparant le mieux les deux classes. La variance du signal filtré sert ensuite de caractéristique à un classifieur linéaire (LDA). La précision rapportée est la proportion d'essais correctement classés,

$$\text{Précision} = \frac{\text{essais bien classés}}{\text{essais totaux}},$$

moyennée sur plusieurs tirages aléatoires indépendants d'entraînement/test (80 %/20 %) pour lisser la variabilité d'un tirage à l'autre.

1.2 MSE ICA vs ICLabel

Pour comparer deux méthodes de débruitage entre elles, on calcule l'erreur quadratique moyenne (MSE) entre les signaux nettoyés correspondants, essai par essai :

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2,$$

où x_i et y_i sont les échantillons (tous canaux et instants confondus) des deux signaux nettoyés et N leur nombre total. Seuls les essais communs aux deux signaux (gauche/droite, repos exclu) sont comparés.

1.3 Évolution du MSE pendant l'entraînement (LOSO)

Le modèle de débruitage est évalué en *leave-one-subject-out* : pour chaque sujet, un modèle neuf est entraîné uniquement sur les autres sujets, puis évalué (même MSE que ci-dessus) sur le sujet exclu, jamais vu pendant l'entraînement. L'opération est répétée pour chaque sujet.

Pendant l'entraînement, on suit deux courbes : le MSE sur les données d'entraînement et le MSE sur le sujet exclu, après chaque passage complet sur les données (epoch). Un écart croissant entre les deux signale un sur-apprentissage.

2 Sujets

2.1 Sujet S001

Signal	Accuracy
Brut	0.63 ± 0.21
ART	0.60 ± 0.15
ICUNet	0.51 ± 0.14
ICUNet++	0.42 ± 0.16
ICUNet_attn	0.49 ± 0.17
DuoCL	0.60 ± 0.11
GCTNet	0.62 ± 0.07
ICLabel	0.67 ± 0.13

TABLE 1 – S001 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ART	64.30	84.91	
ICUNet	17.16	27.65	
ICUNet++	16.01	25.89	
ICUNet_attn	19.43	23.01	26.5
DuoCL	24.82	16.02	
GCTNet	24.27	12.54	

TABLE 2 – S001 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

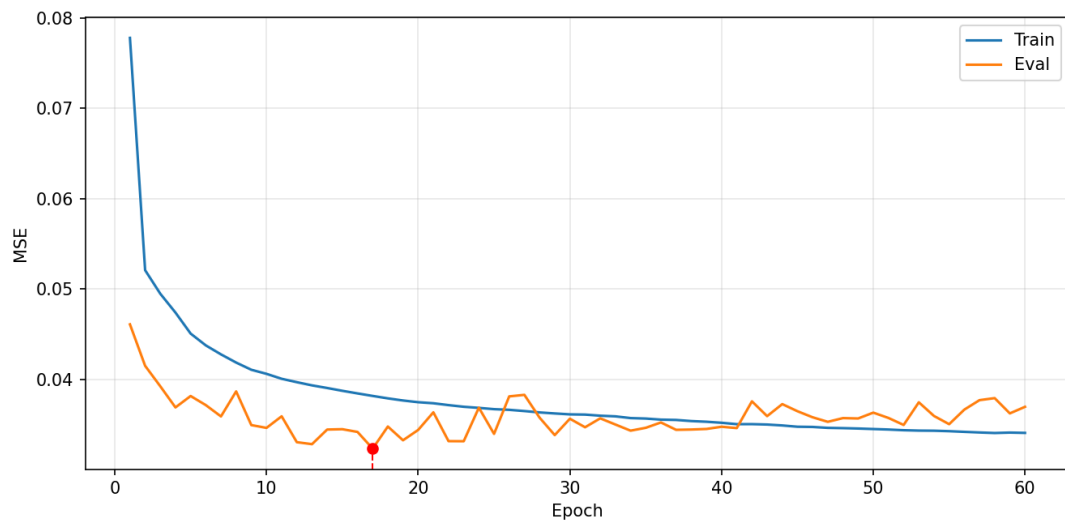


FIGURE 1 – S001 — Évolution du MSE.

Le MSE d'évaluation minimal (0.032) se présente à l'époch 17.

Sujet S001 — ART par epoch

Epoch	Accuracy	RMSE (μ V, vs ICLabel)
1	0.67 ± 0.14	28.56
2	0.68 ± 0.14	28.63
3	0.70 ± 0.13	27.22
4	0.68 ± 0.18	28.52
5	0.69 ± 0.15	25.97
6	0.64 ± 0.12	25.65
7	0.67 ± 0.13	26.57
8	0.69 ± 0.18	25.88
9	0.68 ± 0.15	26.38
10	0.72 ± 0.19	26.88
11	0.70 ± 0.20	26.37
12	0.68 ± 0.18	26.94
13	0.67 ± 0.17	26.74
14	0.67 ± 0.12	27.38
15	0.67 ± 0.12	26.89
16	0.68 ± 0.20	27.00
17	0.70 ± 0.12	27.28
18	0.70 ± 0.14	26.10
19	0.68 ± 0.13	26.72
20	0.69 ± 0.15	27.66
21	0.69 ± 0.16	26.23
22	0.68 ± 0.12	27.78
23	0.68 ± 0.09	27.99
24	0.71 ± 0.12	28.32
25	0.69 ± 0.12	27.91
26	0.71 ± 0.09	28.43
27	0.64 ± 0.11	27.75
28	0.66 ± 0.13	27.98
29	0.68 ± 0.14	27.25
30	0.70 ± 0.12	27.23
31	0.64 ± 0.13	26.92
32	0.67 ± 0.10	27.60
33	0.63 ± 0.12	27.09
34	0.62 ± 0.16	27.12
35	0.64 ± 0.13	26.90
36	0.70 ± 0.12	27.51
37	0.66 ± 0.18	27.01
38	0.63 ± 0.16	27.34
39	0.67 ± 0.13	27.26
40	0.59 ± 0.16	27.44
41	0.68 ± 0.10	26.99
42	0.71 ± 0.12	27.18
43	0.69 ± 0.16	26.92
44	0.68 ± 0.14	28.11
45	0.68 ± 0.13	27.82
46	0.67 ± 0.14	27.72
47	0.64 ± 0.15	27.53
48	0.64 ± 0.13	27.74
49	0.70 ± 0.12	27.30
50	0.68 ± 0.08	27.19
51	0.68 ± 0.10	27.27
52	0.68 ± 0.14	27.48
53	0.67 ± 0.07	28.12
54	0.64 ± 0.13	28.40
55	0.68 ± 0.15	27.07
56	0.71 ± 0.17	28.42
57	0.66 ± 0.08	27.86
58	0.70 ± 0.14	26.88
59	0.67 ± 0.11	27.72
60	0.69 ± 0.08	27.71

Le RMS de référence (signal nettoyé par ICLabel) pour S001 est de 26.5 μ V.

Sujet S001 - ART : Evolution des critères d'évaluation

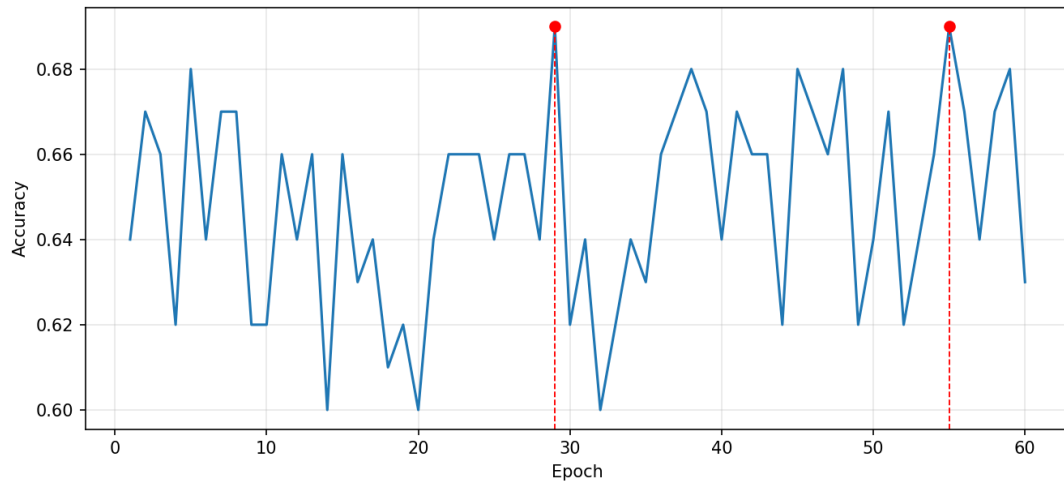


FIGURE 2 – S001 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.69) se présente à l'epoch 29 et à l'epoch 55.

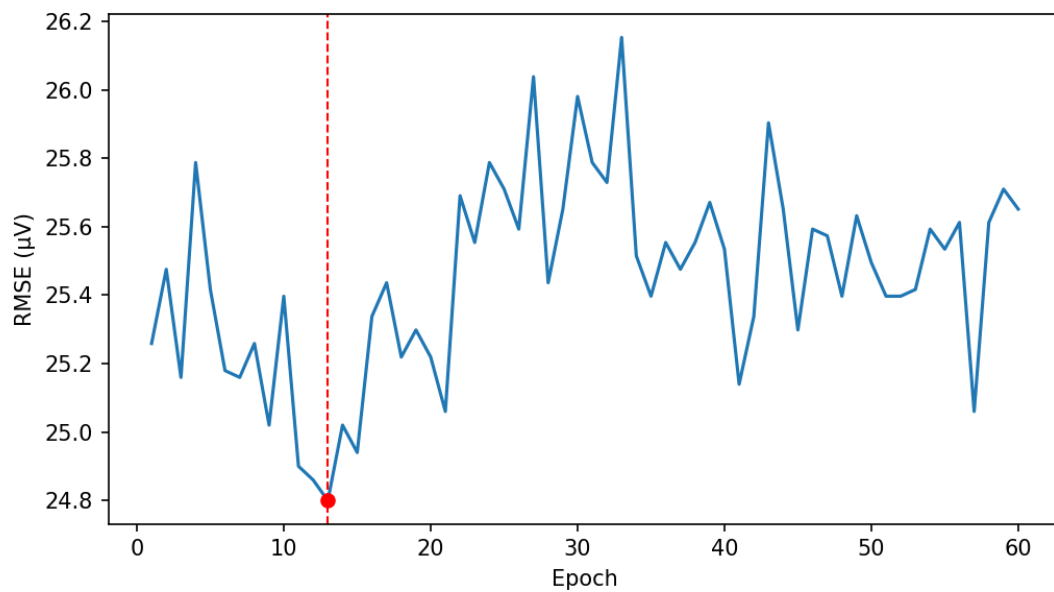


FIGURE 3 – S001 — Évolution du RMSE par epoch (ART vs ICLabel, checkpoints LOSO).

Le RMSE minimal (24.8 μV) se présente à l'epoch 13.

2.2 Sujet S002

Signal	Accuracy
Brut	0.66 ± 0.18
ART	0.66 ± 0.18
ICUNet	0.66 ± 0.14
ICUNet++	0.54 ± 0.20
ICUNet_attn	0.64 ± 0.13
DuoCL	0.67 ± 0.12
GCTNet	0.68 ± 0.14
ICLabel	0.70 ± 0.19

TABLE 3 – S002 — Accuracy de classification.

Comparaison	RMSE (μV)	RMS_Méthode	RMS_ICLABEL
ICUNet	7.14	11.77	
ICUNet++	6.68	12.03	
ICUNet_attn	9.55	12.07	14.7
DuoCL	12.12	9.17	
GCTNet	11.75	7.75	

TABLE 4 – S002 — Erreur (RMSE, μV) par rapport au signal nettoyé par ICLabel.

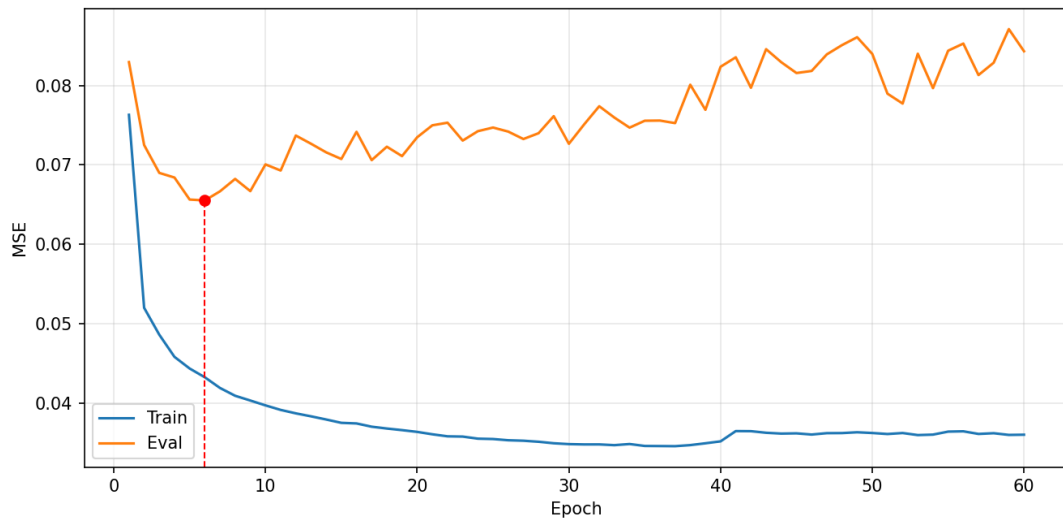


FIGURE 4 – S002 — Évolution du MSE.

Le MSE d'évaluation minimal (0.066) se présente à l'époch 6.

Sujet S002 — ART par epoch

Epoch	Accuracy	RMSE (μV , vs ICLabel)
1	0.72 ± 0.19	13.40
2	0.77 ± 0.18	13.15
3	0.69 ± 0.20	13.17
4	0.69 ± 0.20	13.10
5	0.73 ± 0.19	13.57
6	0.69 ± 0.13	13.32
7	0.69 ± 0.14	13.45
8	0.71 ± 0.15	13.32
9	0.68 ± 0.13	13.52
10	0.70 ± 0.17	13.26
11	0.71 ± 0.19	13.39
12	0.72 ± 0.19	13.30
13	0.72 ± 0.17	13.27
14	0.68 ± 0.17	13.11
15	0.66 ± 0.17	13.34
16	0.64 ± 0.18	12.86
17	0.64 ± 0.18	13.36
18	0.68 ± 0.20	12.42
19	0.67 ± 0.21	13.28
20	0.64 ± 0.21	13.33
21	0.64 ± 0.20	13.38
22	0.70 ± 0.19	13.22
23	0.68 ± 0.22	13.10
24	0.68 ± 0.20	12.99
25	0.66 ± 0.18	13.32
26	0.69 ± 0.21	13.04
27	0.64 ± 0.18	13.40
28	0.70 ± 0.21	13.35
29	0.66 ± 0.18	13.40
30	0.71 ± 0.21	13.30
31	0.69 ± 0.20	13.37
32	0.64 ± 0.17	13.01
33	0.70 ± 0.17	13.24
34	0.66 ± 0.18	13.24
35	0.70 ± 0.17	13.31
36	0.73 ± 0.22	13.41
37	0.70 ± 0.12	13.50
38	0.70 ± 0.18	13.17
39	0.68 ± 0.15	13.63
40	0.70 ± 0.20	13.13
41	0.68 ± 0.18	13.40
42	0.71 ± 0.17	13.16
43	0.66 ± 0.16	12.97
44	0.69 ± 0.21	13.65
45	0.68 ± 0.18	13.09
46	0.71 ± 0.13	13.03
47	0.72 ± 0.18	13.11
48	0.63 ± 0.17	13.30
49	0.66 ± 0.18	13.02
50	0.68 ± 0.19	13.34
51	0.70 ± 0.16	13.27
52	0.69 ± 0.18	13.33
53	0.71 ± 0.19	13.15
54	0.69 ± 0.15	13.33
55	0.69 ± 0.15	13.17
56	0.68 ± 0.16	12.99
57	0.70 ± 0.14	13.07
58	0.70 ± 0.14	13.30
59	0.67 ± 0.17	13.01
60	0.69 ± 0.14	13.00

Le RMS de référence (signal nettoyé par ICLabel) pour S002 est de $14.7 \mu\text{V}$.

Sujet S002 - ART : Evolution des critères d'évaluation

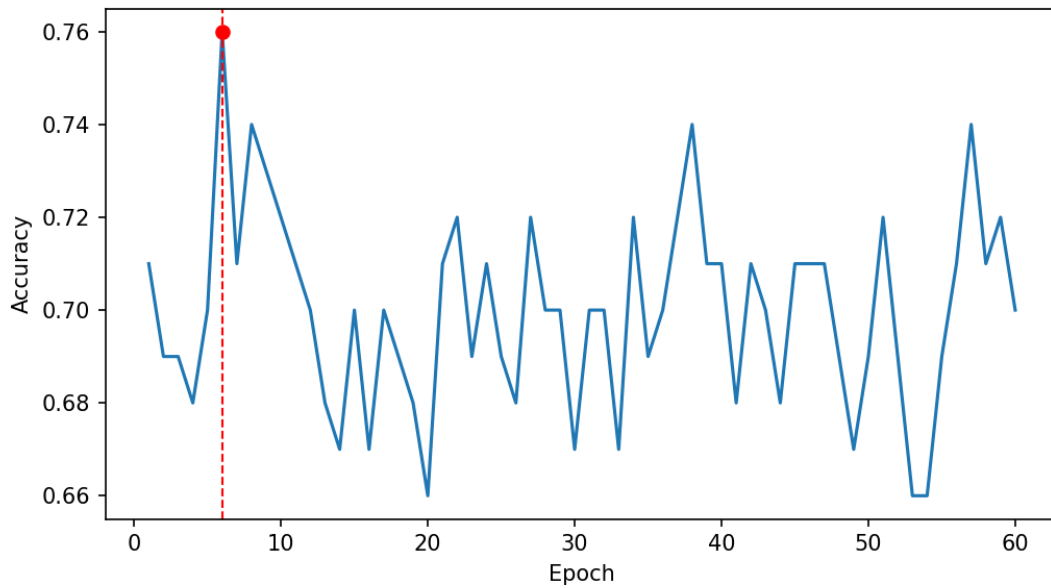


FIGURE 5 – S002 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.76) se présente à l'epoch 6.

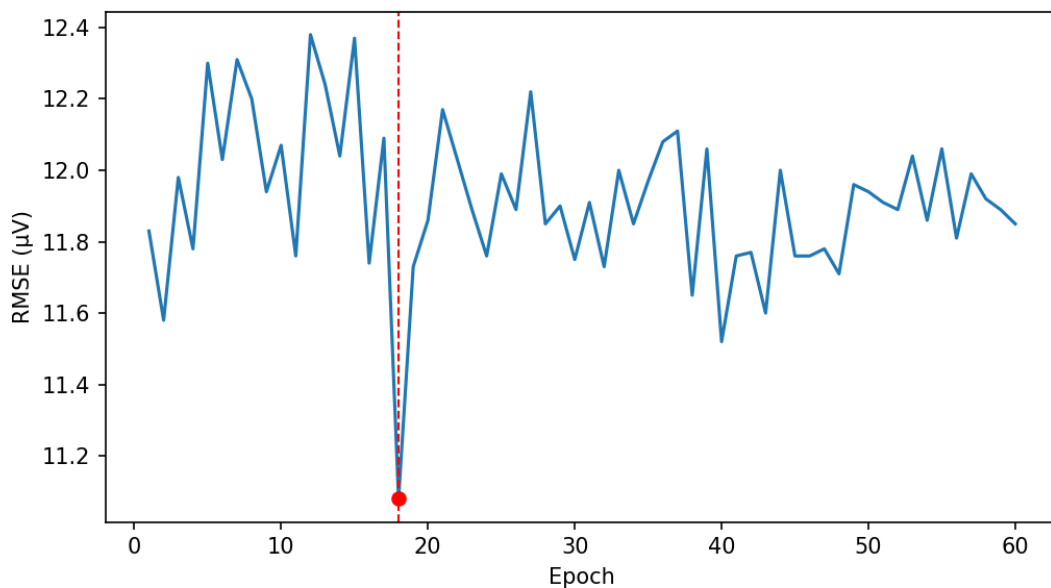


FIGURE 6 – S002 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (11.1 μV) se présente à l'epoch 18.

2.3 Sujet S003

Signal	Accuracy
Brut	0.56 ± 0.16
ICUNet	0.43 ± 0.15
ICUNet++	0.44 ± 0.11
ICUNet_attn	0.50 ± 0.06
DuoCL	0.63 ± 0.12
GCTNet	0.46 ± 0.17
ICLabel	0.66 ± 0.13

TABLE 5 – S003 — Accuracy de classification.

Comparaison	RMSE (μV)	RMS_Méthode	RMS_ICLABEL
ICUNet	21.51	26.67	
ICUNet++	18.88	19.68	
ICUNet_attn	23.69	19.69	25.1
DuoCL	24.52	17.49	
GCTNet	23.60	14.51	

TABLE 6 – S003 — Erreur (RMSE, μV) par rapport au signal nettoyé par ICLabel.

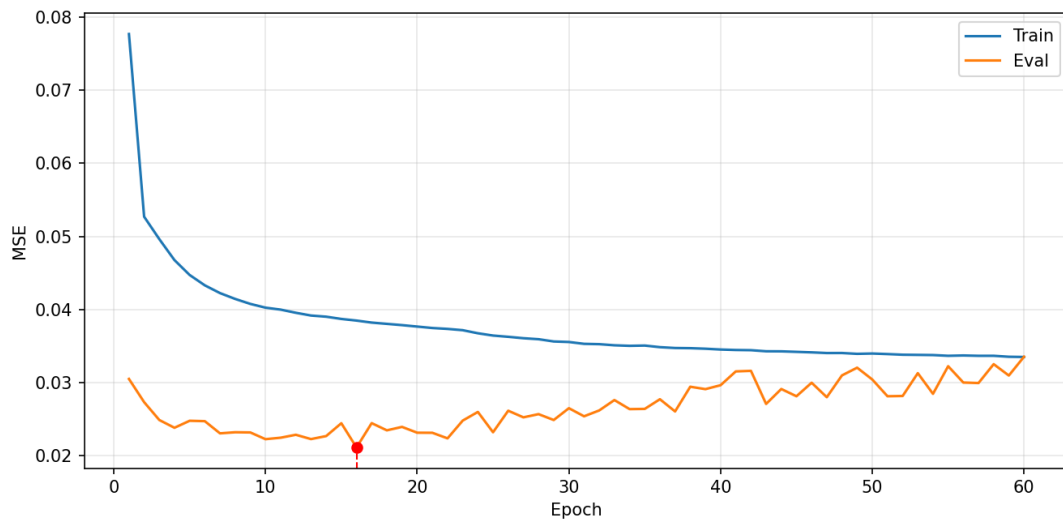


FIGURE 7 – S003 — Évolution du MSE.

Le MSE d'évaluation minimal (0.021) se présente à l'époch 16.

Sujet S003 — ART par epoch

Epoch	Accuracy	RMSE (μ V, vs ICLabel)
1	0.51 ± 0.17	50.67
2	0.44 ± 0.13	47.82
3	0.54 ± 0.17	48.68
4	0.51 ± 0.13	48.15
5	0.52 ± 0.16	43.85
6	0.54 ± 0.17	47.50
7	0.56 ± 0.18	48.33
8	0.53 ± 0.11	48.73
9	0.43 ± 0.15	49.51
10	0.51 ± 0.15	50.43
11	0.53 ± 0.12	48.05
12	0.52 ± 0.16	46.54
13	0.52 ± 0.17	49.86
14	0.54 ± 0.13	48.71
15	0.49 ± 0.17	47.48
16	0.43 ± 0.12	49.94
17	0.42 ± 0.15	48.90
18	0.48 ± 0.13	48.36
19	0.54 ± 0.19	49.03
20	0.50 ± 0.21	46.65
21	0.56 ± 0.19	47.88
22	0.48 ± 0.17	47.11
23	0.52 ± 0.16	49.45
24	0.46 ± 0.16	48.71
25	0.51 ± 0.17	47.85
26	0.52 ± 0.22	46.97
27	0.50 ± 0.12	47.73
28	0.53 ± 0.10	45.04
29	0.47 ± 0.16	48.31
30	0.43 ± 0.15	44.25
31	0.48 ± 0.11	46.30
32	0.47 ± 0.16	45.41
33	0.50 ± 0.13	44.78
34	0.43 ± 0.15	47.42
35	0.50 ± 0.15	45.32
36	0.56 ± 0.16	46.59
37	0.54 ± 0.14	43.58
38	0.51 ± 0.11	45.17
39	0.59 ± 0.16	45.30
40	0.49 ± 0.12	44.15
41	0.49 ± 0.17	46.66
42	0.58 ± 0.14	44.28
43	0.51 ± 0.09	45.13
44	0.47 ± 0.13	45.22
45	0.52 ± 0.12	44.18
46	0.47 ± 0.21	49.24
47	0.51 ± 0.14	46.87
48	0.42 ± 0.15	45.71
49	0.53 ± 0.11	44.75
50	0.47 ± 0.20	43.86
51	0.48 ± 0.17	44.49
52	0.50 ± 0.12	44.14
53	0.58 ± 0.13	43.77
54	0.51 ± 0.19	46.30
55	0.49 ± 0.17	44.14
56	0.50 ± 0.13	42.76
57	0.54 ± 0.18	46.63
58	0.53 ± 0.16	43.52
59	0.52 ± 0.19	45.43
60	0.48 ± 0.10	44.67

Le RMS de référence (signal nettoyé par ICLabel) pour S003 est de 25.1 μ V.

Sujet S003 - ART : Evolution des critères d'évaluation

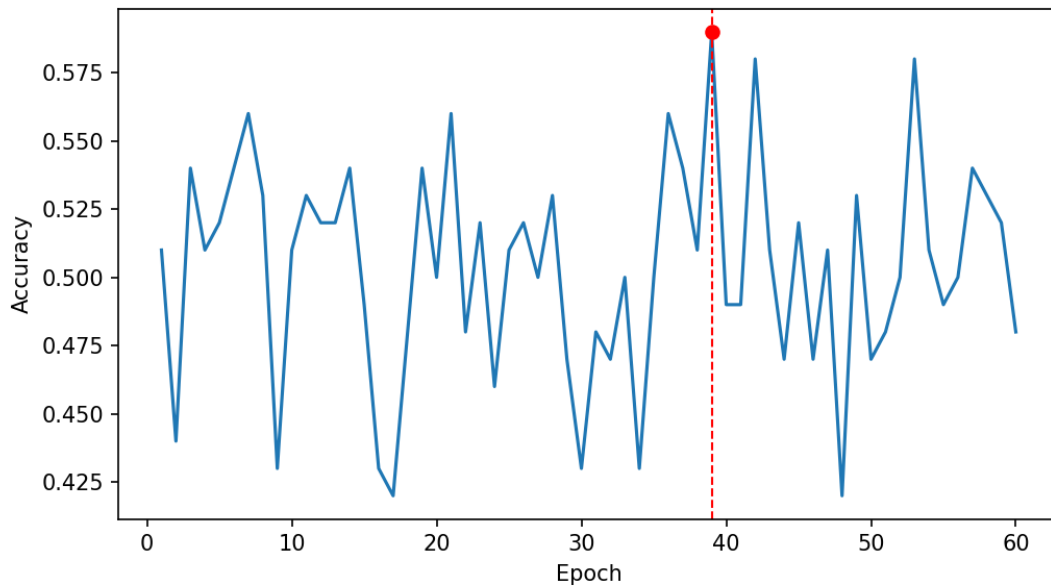


FIGURE 8 – S003 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.59) se présente à l'epoch 39.

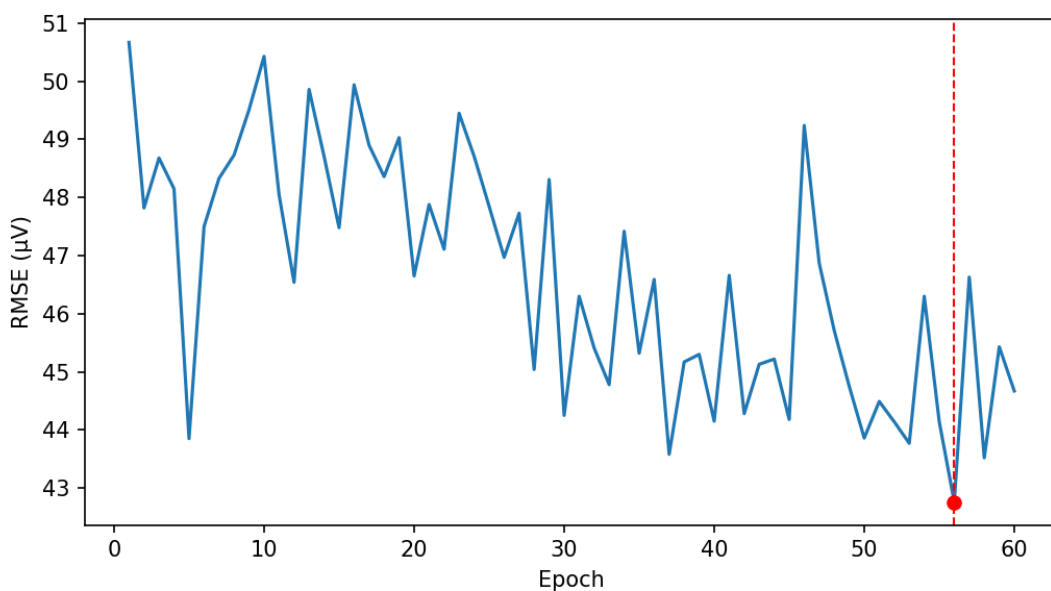


FIGURE 9 – S003 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (42.8 μV) se présente à l'epoch 56.

2.4 Sujet S004

Signal	Accuracy
Brut	0.43 ± 0.16
ICUNet	0.58 ± 0.17
ICUNet++	0.66 ± 0.13
ICUNet_attn	0.36 ± 0.15
DuoCL	0.60 ± 0.19
GCTNet	0.54 ± 0.12
ICLabel	0.44 ± 0.11

TABLE 7 – S004 — Accuracy de classification.

Comparaison	RMSE (μV)	RMS_Méthode	RMS_ICLABEL
ICUNet	23.58	12.21	
ICUNet++	25.08	7.53	
ICUNet_attn	26.24	10.50	26.1
DuoCL	23.81	8.72	
GCTNet	24.33	7.30	

TABLE 8 – S004 — Erreur (RMSE, μV) par rapport au signal nettoyé par ICLabel.

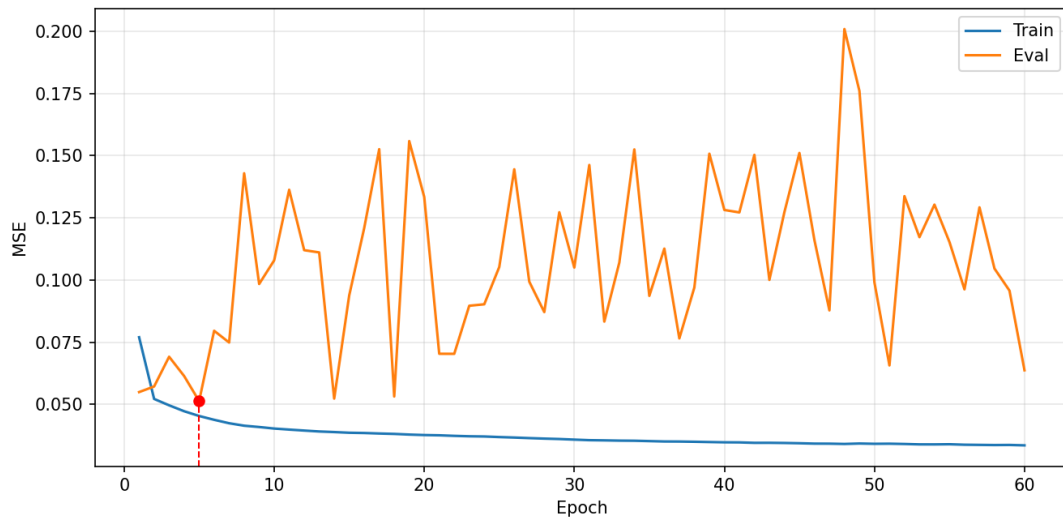


FIGURE 10 – S004 — Évolution du MSE.

Le MSE d'évaluation minimal (0.051) se présente à l'époch 5.

Sujet S004 — ART par epoch

Epoch	Accuracy	RMSE (μV , vs ICLabel)
1	0.41 ± 0.16	21.82
2	0.46 ± 0.14	21.18
3	0.51 ± 0.15	21.32
4	0.38 ± 0.12	19.69
5	0.48 ± 0.18	19.28
6	0.48 ± 0.16	19.40
7	0.42 ± 0.16	20.08
8	0.43 ± 0.16	21.25
9	0.44 ± 0.15	20.45
10	0.51 ± 0.17	21.94
11	0.51 ± 0.17	21.17
12	0.48 ± 0.20	20.93
13	0.48 ± 0.13	20.80
14	0.51 ± 0.22	19.11
15	0.49 ± 0.19	21.40
16	0.48 ± 0.17	20.97
17	0.43 ± 0.19	22.66
18	0.49 ± 0.18	20.54
19	0.53 ± 0.20	22.67
20	0.52 ± 0.13	23.05
21	0.44 ± 0.21	22.00
22	0.44 ± 0.12	21.59
23	0.40 ± 0.14	22.07
24	0.48 ± 0.10	20.68
25	0.54 ± 0.12	22.31
26	0.42 ± 0.15	22.71
27	0.42 ± 0.13	21.56
28	0.56 ± 0.18	21.73
29	0.42 ± 0.18	22.59
30	0.42 ± 0.23	22.48
31	0.41 ± 0.14	21.70
32	0.48 ± 0.14	20.40
33	0.43 ± 0.14	22.05
34	0.53 ± 0.13	21.60
35	0.48 ± 0.11	21.94
36	0.38 ± 0.13	21.30
37	0.38 ± 0.10	20.56
38	0.52 ± 0.12	21.79
39	0.47 ± 0.16	23.36
40	0.70 ± 0.10	21.90
41	0.50 ± 0.10	21.85
42	0.47 ± 0.12	22.47
43	0.52 ± 0.20	21.42
44	0.52 ± 0.16	21.85
45	0.56 ± 0.13	22.46
46	0.48 ± 0.09	20.87
47	0.51 ± 0.09	20.71
48	0.46 ± 0.13	22.73
49	0.41 ± 0.14	22.72
50	0.48 ± 0.19	20.52
51	0.37 ± 0.17	21.17
52	0.52 ± 0.14	21.30
53	0.50 ± 0.11	21.74
54	0.46 ± 0.15	21.23
55	0.42 ± 0.15	21.59
56	0.51 ± 0.17	20.87
57	0.58 ± 0.22	20.56
58	0.54 ± 0.14	20.70
59	0.44 ± 0.12	20.86
60	0.60 ± 0.10	20.63

Le RMS de référence (signal nettoyé par ICLabel) pour S004 est de $26.1 \mu\text{V}$.

Sujet S004 - ART : Evolution des critères d'évaluation

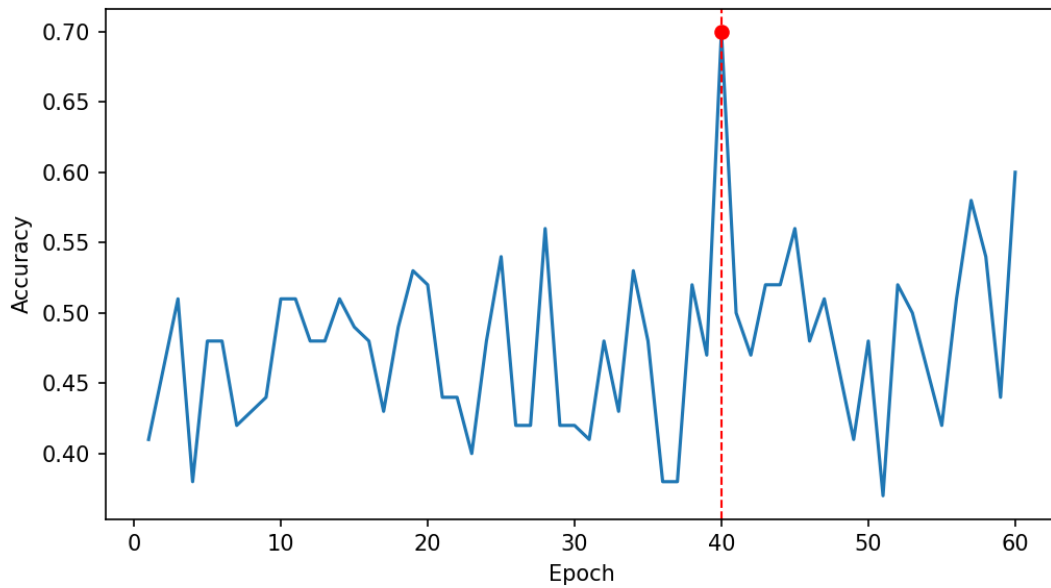


FIGURE 11 – S004 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.70) se présente à l'epoch 40.

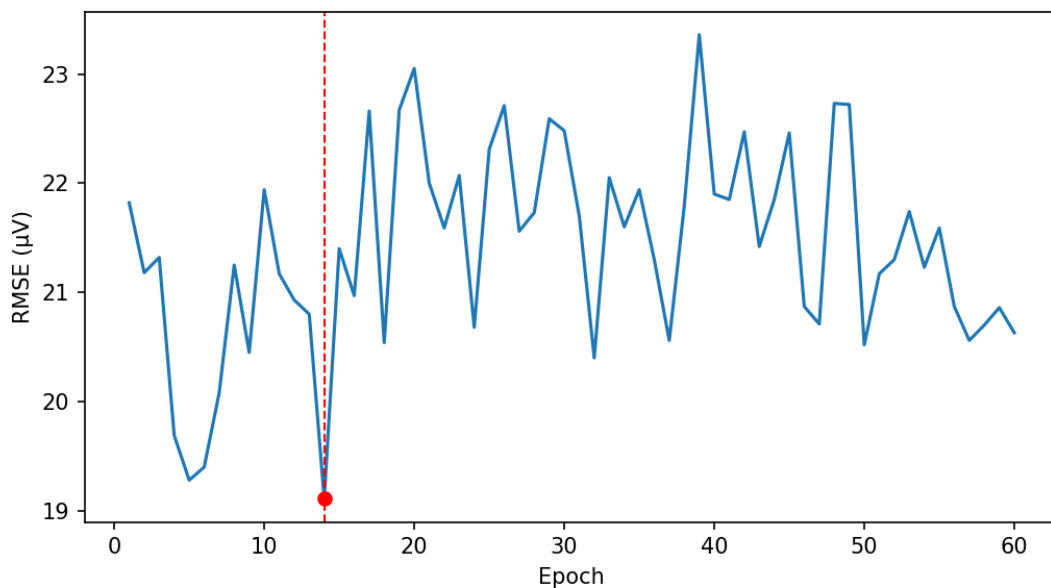


FIGURE 12 – S004 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (19.1 μV) se présente à l'epoch 14.

2.5 Sujet S005

Signal	Accuracy
Brut	0.49 ± 0.14
ICUNet	0.48 ± 0.14
ICUNet++	0.44 ± 0.16
ICUNet_attn	0.33 ± 0.16
DuoCL	0.37 ± 0.17
GCTNet	0.60 ± 0.10
ICLabel	0.49 ± 0.12

TABLE 9 – S005 — Accuracy de classification.

Comparaison	RMSE (μV)	RMS_Méthode	RMS_ICLABEL
ICUNet	12.13	9.05	
ICUNet++	12.64	8.55	
ICUNet_attn	14.66	7.83	16.7
DuoCL	14.80	8.48	
GCTNet	14.59	7.01	

TABLE 10 – S005 — Erreur (RMSE, μV) par rapport au signal nettoyé par ICLabel.

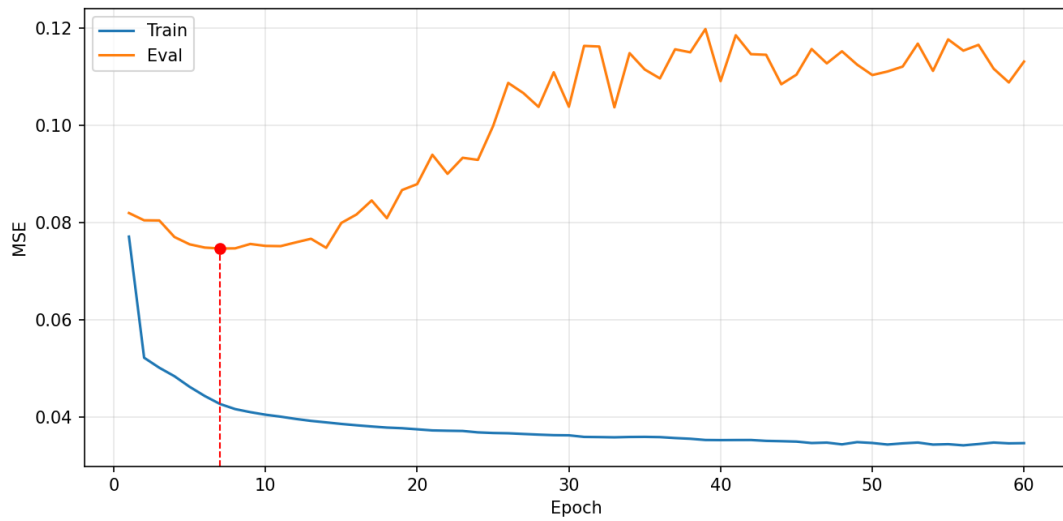


FIGURE 13 – S005 — Évolution du MSE.

Le MSE d'évaluation minimal (0.075) se présente à l'époch 7.

Sujet S005 — ART par epoch

Epoch	Accuracy	RMSE (μV , vs ICLabel)
1	0.52 ± 0.13	16.98
2	0.51 ± 0.15	16.09
3	0.53 ± 0.20	16.17
4	0.54 ± 0.14	15.61
5	0.52 ± 0.11	16.20
6	0.51 ± 0.16	16.30
7	0.47 ± 0.10	16.14
8	0.52 ± 0.17	16.08
9	0.50 ± 0.13	15.20
10	0.49 ± 0.14	15.18
11	0.58 ± 0.17	15.81
12	0.52 ± 0.19	15.85
13	0.50 ± 0.16	16.34
14	0.51 ± 0.17	15.74
15	0.54 ± 0.17	14.76
16	0.52 ± 0.19	14.51
17	0.54 ± 0.08	14.69
18	0.54 ± 0.10	14.94
19	0.54 ± 0.10	14.15
20	0.49 ± 0.14	14.32
21	0.51 ± 0.12	14.05
22	0.50 ± 0.14	13.86
23	0.50 ± 0.14	14.55
24	0.53 ± 0.13	14.17
25	0.53 ± 0.11	13.91
26	0.50 ± 0.06	13.66
27	0.48 ± 0.14	13.40
28	0.53 ± 0.11	13.75
29	0.48 ± 0.12	13.69
30	0.52 ± 0.10	13.91
31	0.47 ± 0.10	13.25
32	0.50 ± 0.12	13.30
33	0.52 ± 0.10	13.88
34	0.49 ± 0.10	13.71
35	0.49 ± 0.12	13.69
36	0.48 ± 0.12	13.49
37	0.47 ± 0.12	13.35
38	0.53 ± 0.11	13.35
39	0.52 ± 0.11	13.09
40	0.54 ± 0.13	13.54
41	0.51 ± 0.14	13.28
42	0.52 ± 0.09	13.86
43	0.52 ± 0.07	13.57
44	0.50 ± 0.12	13.84
45	0.48 ± 0.11	13.63
46	0.58 ± 0.08	13.82
47	0.52 ± 0.14	13.48
48	0.52 ± 0.07	14.04
49	0.46 ± 0.12	13.39
50	0.53 ± 0.14	13.53
51	0.49 ± 0.14	13.47
52	0.50 ± 0.10	13.70
53	0.52 ± 0.10	13.19
54	0.53 ± 0.15	13.93
55	0.51 ± 0.12	13.72
56	0.52 ± 0.09	13.47
57	0.52 ± 0.13	13.55
58	0.54 ± 0.09	13.58
59	0.52 ± 0.10	13.63
60	0.50 ± 0.11	13.47

Le RMS de référence (signal nettoyé par ICLabel) pour S005 est de $16.7 \mu\text{V}$.

Sujet S005 - ART : Evolution des critères d'évaluation

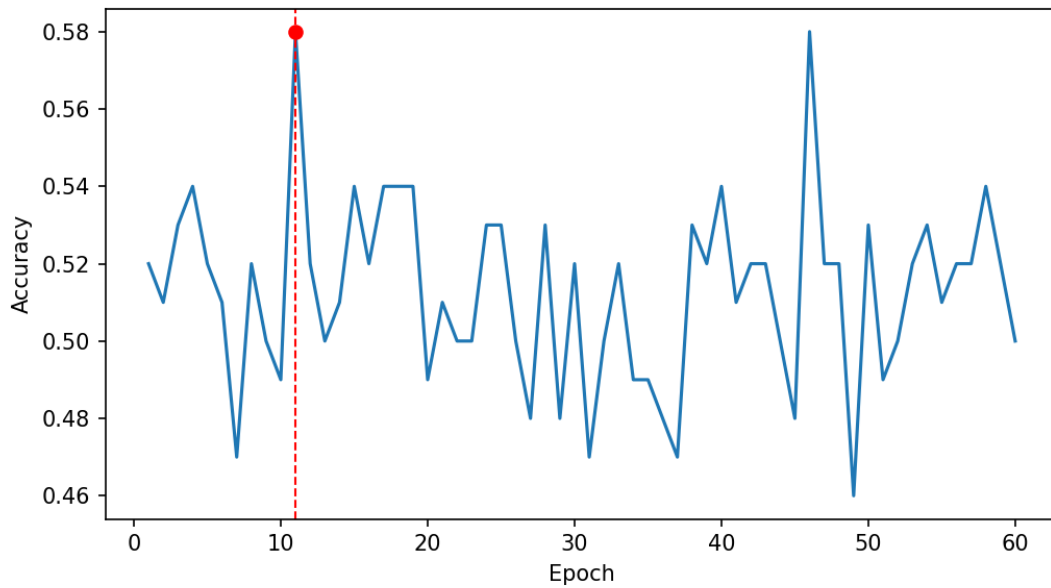


FIGURE 14 – S005 — Évolution de l'accuracy par epoch (ART, checkpoints LOSO).

La meilleure accuracy (0.58) se présente à l'epoch 11.

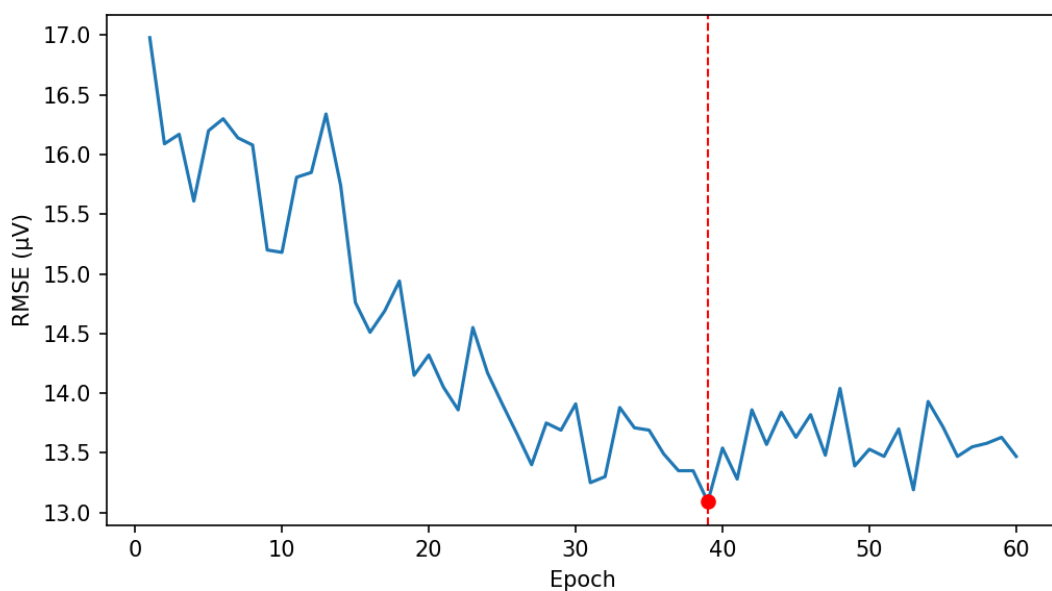


FIGURE 15 – S005 — Évolution du RMSE par epoch (ART vs ICLLabel, checkpoints LOSO).

Le RMSE minimal (13.1 μV) se présente à l'epoch 39.

2.6 Sujet S006

Signal	Accuracy
Brut	0.50 ± 0.11
ICUNet	0.53 ± 0.12
ICUNet++	0.49 ± 0.18
ICUNet_attn	0.57 ± 0.18
DuoCL	0.52 ± 0.09
GCTNet	0.47 ± 0.17
ICLabel	0.51 ± 0.15

TABLE 11 – S006 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	8.05	7.56	
ICUNet++	8.33	6.93	
ICUNet_attn	9.89	8.13	11.8
DuoCL	11.05	6.80	
GCTNet	10.91	6.22	

TABLE 12 – S006 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

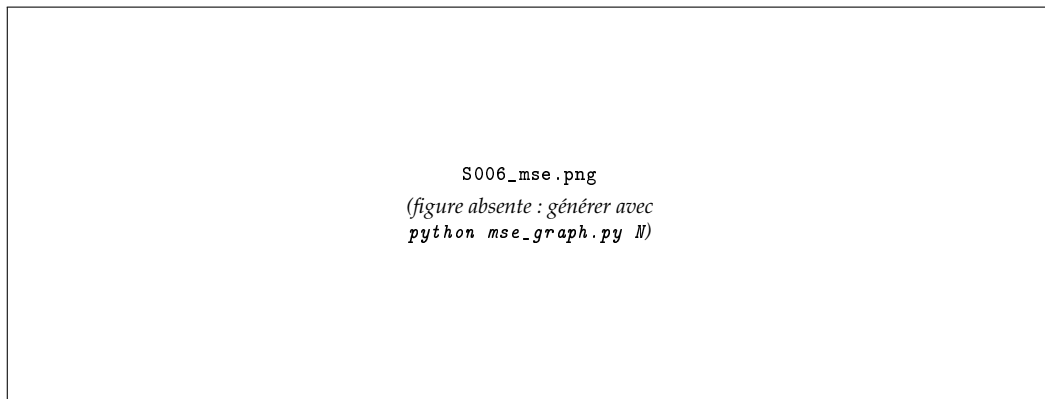


FIGURE 16 – S006 — Évolution du MSE.

2.7 Sujet S007

Signal	Accuracy
Brut	0.96 ± 0.07
ICUNet	0.82 ± 0.10
ICUNet++	0.77 ± 0.09
ICUNet_attn	0.61 ± 0.17
DuoCL	0.81 ± 0.10
GCTNet	0.91 ± 0.08
ICLabel	0.98 ± 0.04

TABLE 13 – S007 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	8.20	19.50	
ICUNet++	7.88	19.25	
ICUNet_attn	13.22	19.46	21.3
DuoCL	15.56	15.70	
GCTNet	15.00	13.48	

TABLE 14 – S007 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

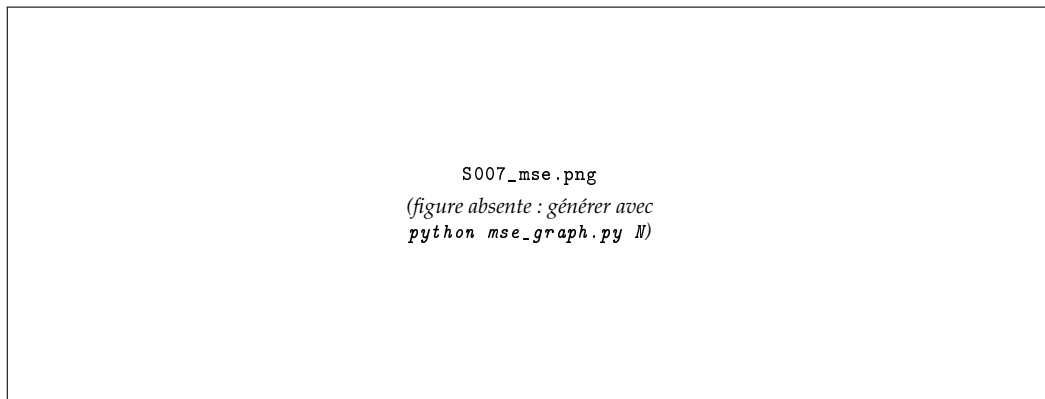


FIGURE 17 – S007 — Évolution du MSE.

2.8 Sujet S008

Signal	Accuracy
Brut	0.48 ± 0.15
ICUNet	0.59 ± 0.11
ICUNet++	0.52 ± 0.15
ICUNet_attn	0.46 ± 0.13
DuoCL	0.42 ± 0.18
GCTNet	0.38 ± 0.15
ICLabel	0.52 ± 0.15

TABLE 15 – S008 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	9.57	13.02	
ICUNet++	10.14	11.23	
ICUNet_attn	12.96	10.53	16.1
DuoCL	14.04	9.52	
GCTNet	13.75	7.75	

TABLE 16 – S008 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

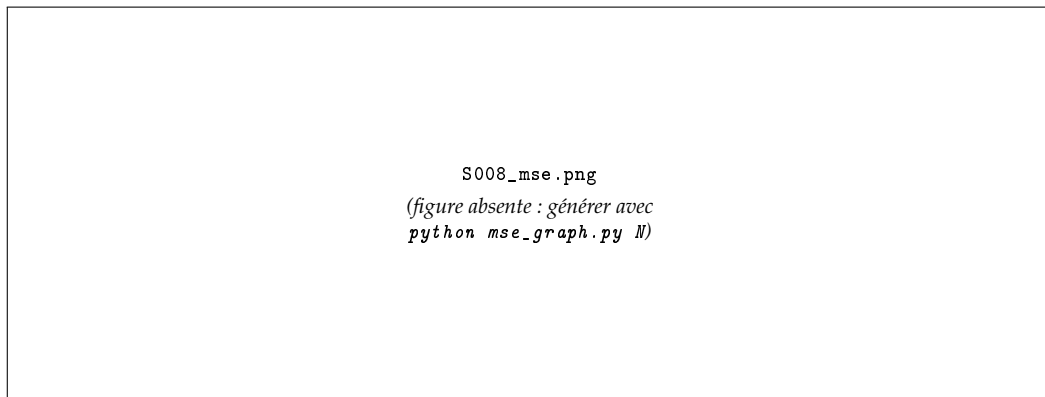


FIGURE 18 – S008 — Évolution du MSE.

2.9 Sujet S009

Signal	Accuracy
Brut	0.38 ± 0.19
ICUNet	0.42 ± 0.17
ICUNet++	0.42 ± 0.16
ICUNet_attn	0.39 ± 0.16
DuoCL	0.29 ± 0.11
GCTNet	0.37 ± 0.17
ICLabel	0.56 ± 0.17

TABLE 17 – S009 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	19.74	17.67	
ICUNet++	19.79	15.67	
ICUNet_attn	25.21	19.55	27.0
DuoCL	25.73	17.79	
GCTNet	25.04	16.05	

TABLE 18 – S009 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

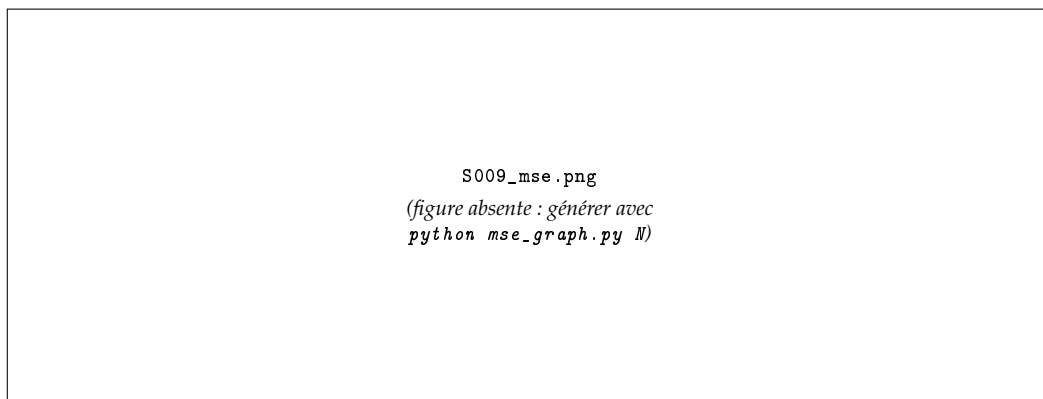


FIGURE 19 – S009 — Évolution du MSE.

2.10 Sujet S010

Signal	Accuracy
Brut	0.58 ± 0.16
ICUNet	0.42 ± 0.11
ICUNet++	0.42 ± 0.13
ICUNet_attn	0.41 ± 0.07
DuoCL	0.59 ± 0.16
GCTNet	0.58 ± 0.11
ICLabel	0.60 ± 0.12

TABLE 19 – S010 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	18.67	25.23	
ICUNet++	18.73	24.23	
ICUNet_attn	22.48	22.89	27.6
DuoCL	25.61	16.38	
GCTNet	25.34	12.70	

TABLE 20 – S010 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

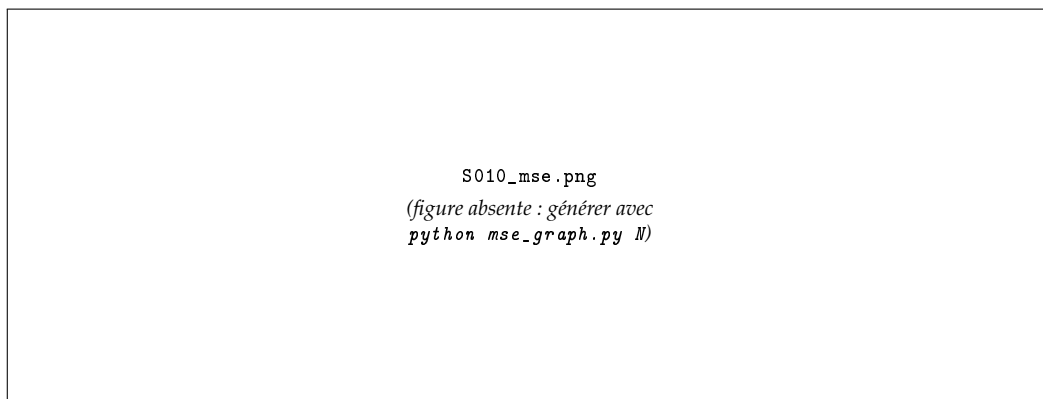


FIGURE 20 – S010 — Évolution du MSE.

2.11 Sujet S011

Signal	Accuracy
Brut	0.42 ± 0.14
ICUNet	0.52 ± 0.15
ICUNet++	0.42 ± 0.18
ICUNet_attn	0.44 ± 0.19
DuoCL	0.44 ± 0.12
GCTNet	0.41 ± 0.16
ICLabel	0.50 ± 0.20

TABLE 21 – S011 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	5.73	10.48	
ICUNet++	6.73	9.09	
ICUNet_attn	9.51	8.06	13.3
DuoCL	11.53	6.32	
GCTNet	11.49	5.27	

TABLE 22 – S011 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

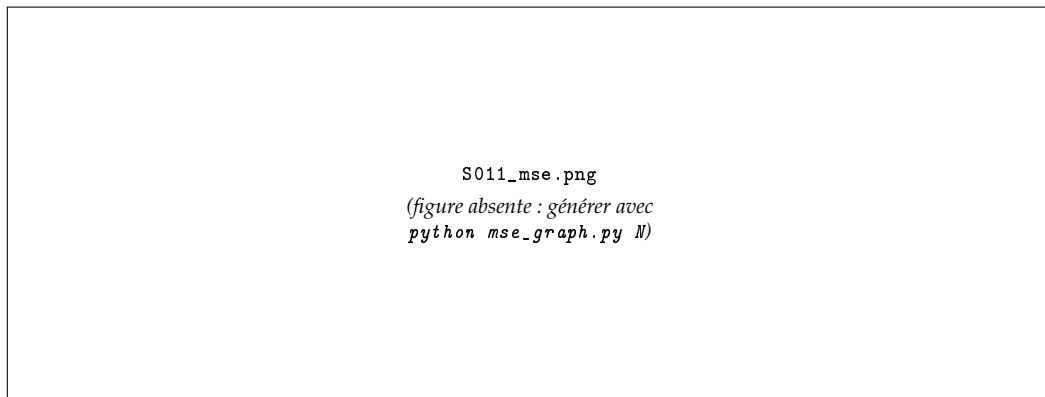


FIGURE 21 – S011 — Évolution du MSE.

2.12 Sujet S012

Signal	Accuracy
Brut	0.53 ± 0.16
ICUNet	0.50 ± 0.12
ICUNet++	0.37 ± 0.09
ICUNet_attn	0.48 ± 0.17
DuoCL	0.59 ± 0.17
GCTNet	0.50 ± 0.16
ICLabel	0.50 ± 0.17

TABLE 23 – S012 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	12.19	15.40	
ICUNet++	10.97	12.50	
ICUNet_attn	14.73	13.43	15.7
DuoCL	14.73	8.41	
GCTNet	14.56	7.05	

TABLE 24 – S012 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

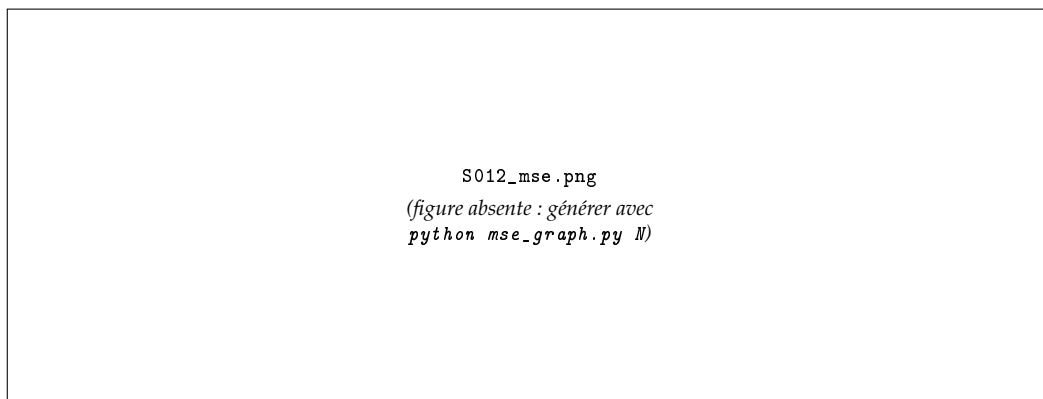


FIGURE 22 – S012 — Évolution du MSE.

2.13 Sujet S013

Signal	Accuracy
Brut	0.50 ± 0.17
ICUNet	0.41 ± 0.17
ICUNet++	0.32 ± 0.14
ICUNet_attn	0.58 ± 0.17
DuoCL	0.42 ± 0.14
GCTNet	0.49 ± 0.16
ICLabel	0.51 ± 0.18

TABLE 25 – S013 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	19.97	21.23	
ICUNet++	20.72	21.08	
ICUNet_attn	25.48	23.50	29.6
DuoCL	27.68	21.10	
GCTNet	27.29	18.26	

TABLE 26 – S013 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

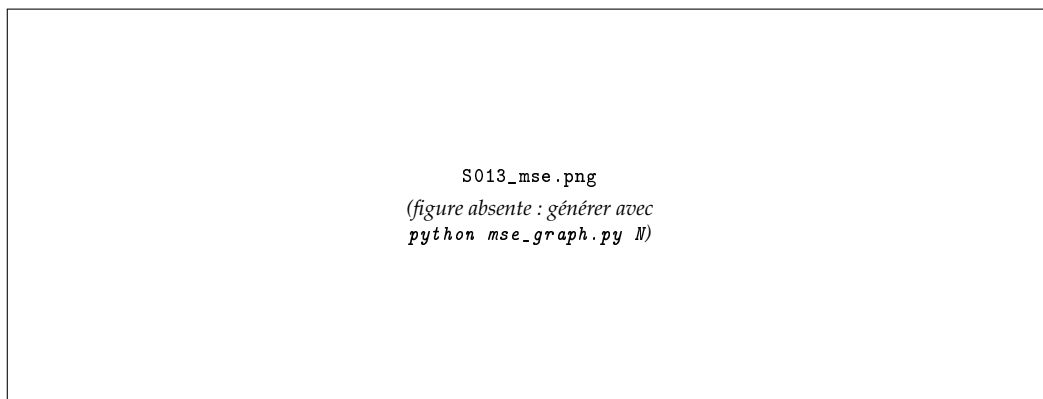


FIGURE 23 – S013 — Évolution du MSE.

2.14 Sujet S014

Signal	Accuracy
Brut	0.47 ± 0.12
ICUNet	0.54 ± 0.12
ICUNet++	0.61 ± 0.11
ICUNet_attn	0.48 ± 0.14
DuoCL	0.38 ± 0.11
GCTNet	0.53 ± 0.13
ICLabel	0.54 ± 0.13

TABLE 27 – S014 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	9.85	16.66	
ICUNet++	9.80	16.98	
ICUNet_attn	13.28	16.83	20.7
DuoCL	16.61	13.84	
GCTNet	16.34	11.03	

TABLE 28 – S014 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

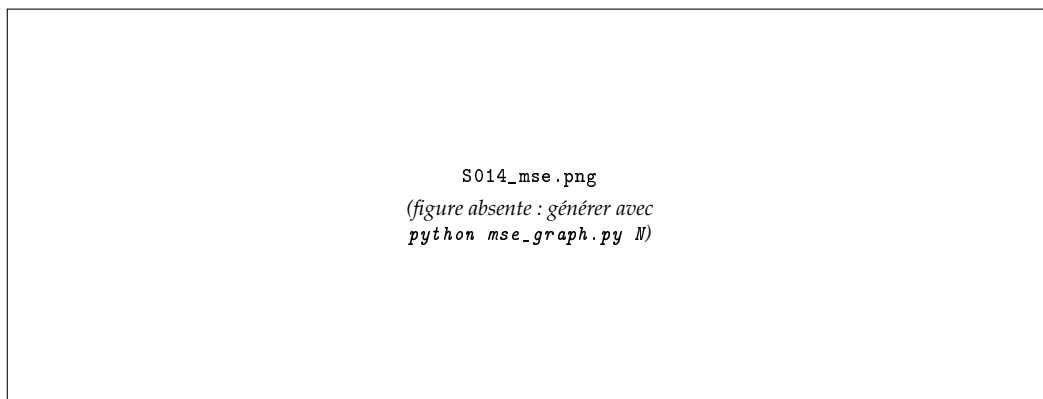


FIGURE 24 – S014 — Évolution du MSE.

2.15 Sujet S015

Signal	Accuracy
Brut	0.66 ± 0.12
ICUNet	0.63 ± 0.16
ICUNet++	0.47 ± 0.14
ICUNet_attn	0.48 ± 0.13
DuoCL	0.63 ± 0.14
GCTNet	0.44 ± 0.19
ICLabel	0.69 ± 0.16

TABLE 29 – S015 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	14.63	27.60	
ICUNet++	16.03	25.33	
ICUNet_attn	22.80	24.58	34.7
DuoCL	28.46	20.62	
GCTNet	27.64	17.06	

TABLE 30 – S015 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

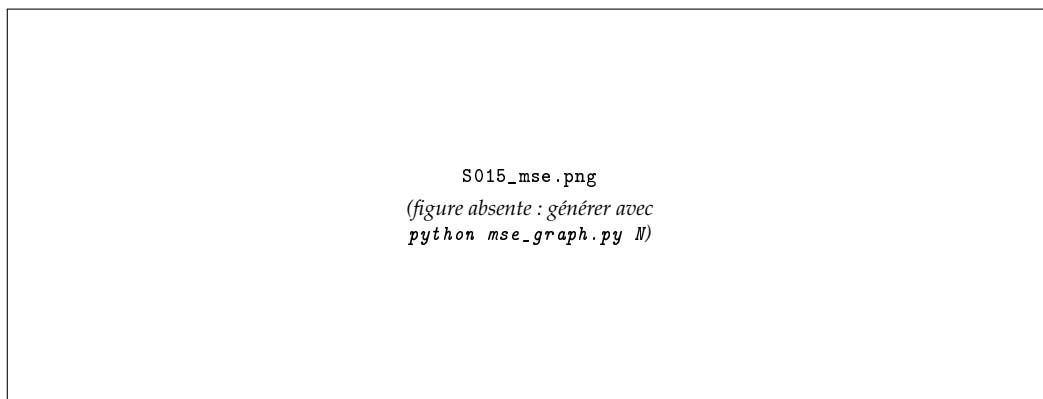


FIGURE 25 – S015 — Évolution du MSE.

2.16 Sujet S016

Signal	Accuracy
Brut	0.47 ± 0.15
ICUNet	0.51 ± 0.09
ICUNet++	0.54 ± 0.14
ICUNet_attn	0.49 ± 0.16
DuoCL	0.60 ± 0.11
GCTNet	0.58 ± 0.16
ICLabel	0.66 ± 0.15

TABLE 31 – S016 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	10.84	14.00	
ICUNet++	9.64	12.24	
ICUNet_attn	11.34	11.16	12.6
DuoCL	12.25	10.08	
GCTNet	11.45	8.10	

TABLE 32 – S016 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

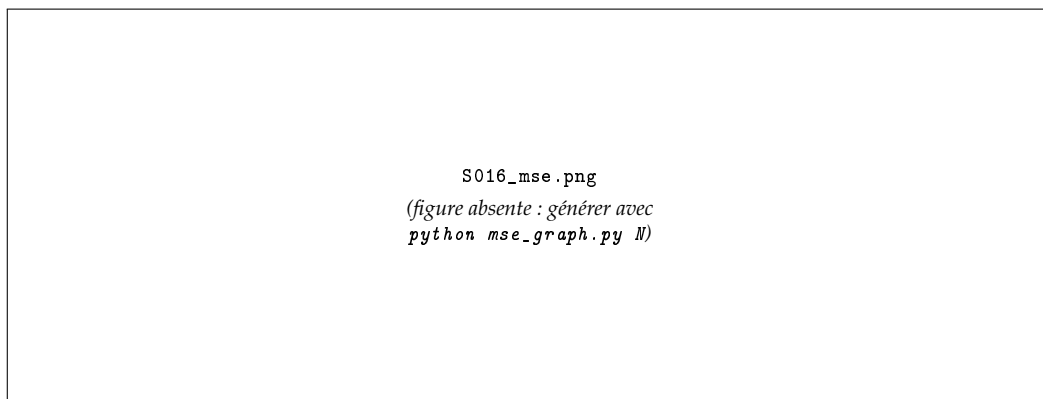


FIGURE 26 – S016 — Évolution du MSE.

2.17 Sujet S017

Signal	Accuracy
Brut	0.40 ± 0.18
ICUNet	0.37 ± 0.20
ICUNet++	0.41 ± 0.07
ICUNet_attn	0.54 ± 0.17
DuoCL	0.52 ± 0.15
GCTNet	0.36 ± 0.16
ICLabel	0.34 ± 0.16

TABLE 33 – S017 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	27.77	22.06	
ICUNet++	28.16	19.09	
ICUNet_attn	33.72	21.18	36.9
DuoCL	34.35	19.61	
GCTNet	33.17	17.73	

TABLE 34 – S017 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

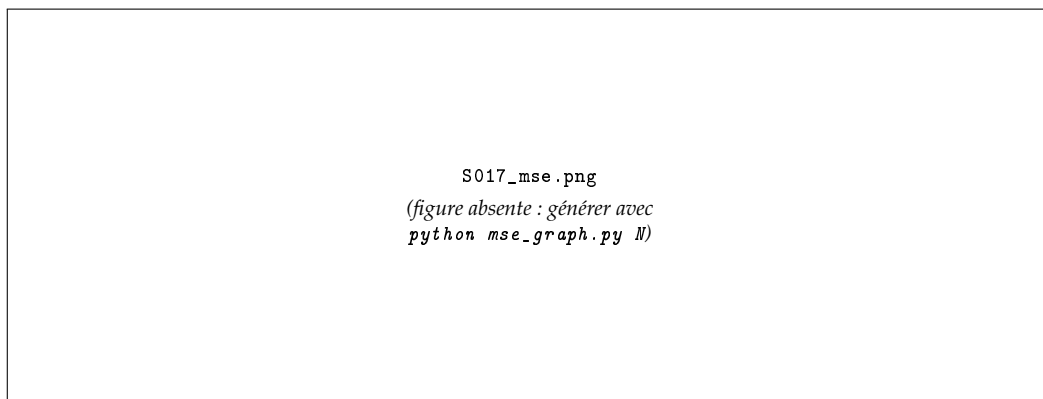


FIGURE 27 – S017 — Évolution du MSE.

2.18 Sujet S018

Signal	Accuracy
Brut	0.39 ± 0.13
ICUNet	0.32 ± 0.10
ICUNet++	0.52 ± 0.17
ICUNet_attn	0.52 ± 0.15
DuoCL	0.49 ± 0.11
GCTNet	0.51 ± 0.17
ICLabel	0.36 ± 0.16

TABLE 35 – S018 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	5.01	6.30	
ICUNet++	4.27	4.99	
ICUNet_attn	5.68	5.32	5.8
DuoCL	5.73	3.83	
GCTNet	5.50	3.15	

TABLE 36 – S018 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

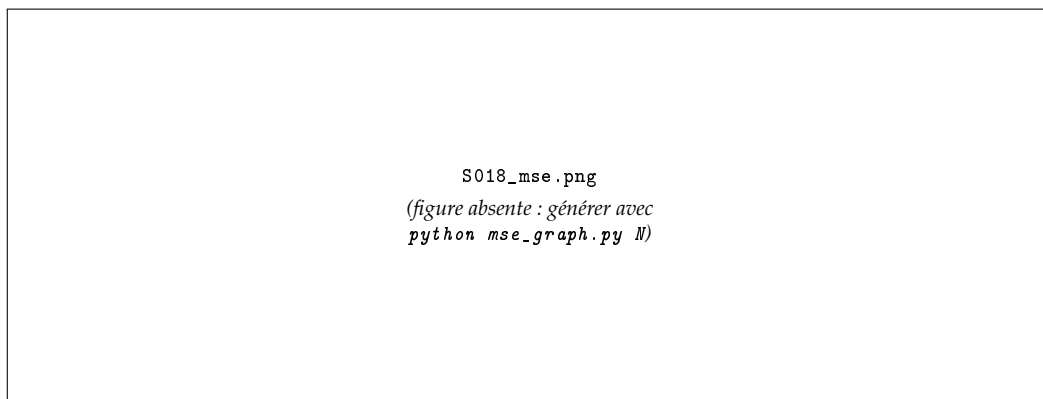


FIGURE 28 – S018 — Évolution du MSE.

2.19 Sujet S019

Signal	Accuracy
Brut	0.60 ± 0.07
ICUNet	0.69 ± 0.13
ICUNet++	0.29 ± 0.16
ICUNet_attn	0.59 ± 0.12
DuoCL	0.62 ± 0.07
GCTNet	0.68 ± 0.12
ICLabel	0.68 ± 0.08

TABLE 37 – S019 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	15.80	15.01	
ICUNet++	16.70	14.04	
ICUNet_attn	20.16	14.38	24.9
DuoCL	21.84	14.07	
GCTNet	21.61	12.49	

TABLE 38 – S019 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

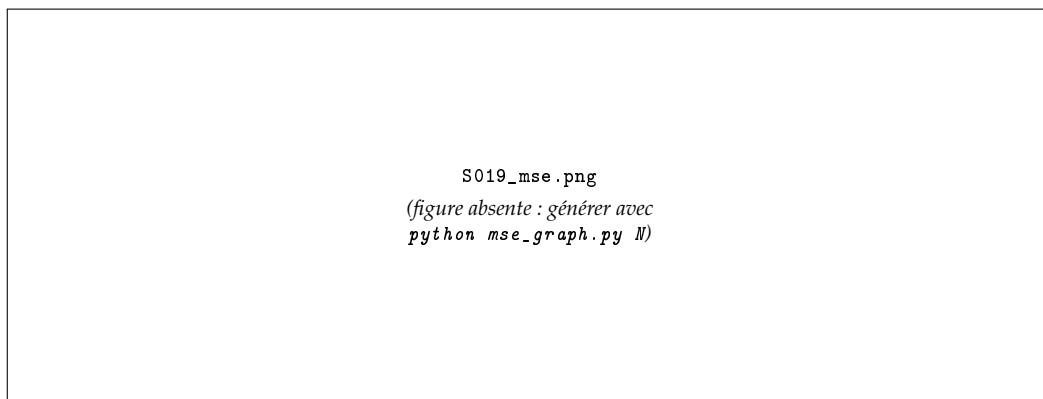


FIGURE 29 – S019 — Évolution du MSE.

2.20 Sujet S020

Signal	Accuracy
Brut	0.62 ± 0.16
ICUNet	0.49 ± 0.17
ICUNet++	0.40 ± 0.23
ICUNet_attn	0.48 ± 0.10
DuoCL	0.39 ± 0.15
GCTNet	0.61 ± 0.11
ICLabel	0.66 ± 0.18

TABLE 39 – S020 — Accuracy de classification.

Comparaison	RMSE (μ V)	RMS_Méthode	RMS_ICLABEL
ICUNet	5.00	5.78	
ICUNet++	4.84	5.28	
ICUNet_attn	5.48	5.05	5.4
DuoCL	5.32	3.59	
GCTNet	5.15	2.88	

TABLE 40 – S020 — Erreur (RMSE, μ V) par rapport au signal nettoyé par ICLabel.

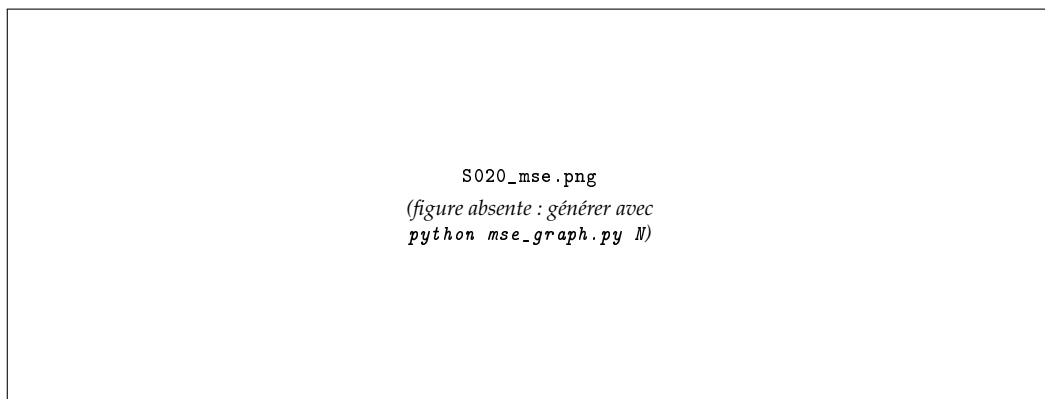


FIGURE 30 – S020 — Évolution du MSE.

3 Récapitulatif

TABLE 41 – Récapitulatif — accuracy CSP+LDA, les 109 sujets, par méthode (« - » = non calculé).

Sujet	Brut	ICLabel	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S001	0.63 ± 0.21	0.67 ± 0.13	0.60 ± 0.11	0.62 ± 0.07	0.51 ± 0.14	0.42 ± 0.16	0.49 ± 0.17
S002	0.68 ± 0.18	0.70 ± 0.19	0.67 ± 0.12	0.68 ± 0.14	0.66 ± 0.14	0.54 ± 0.20	0.64 ± 0.13
S003	0.56 ± 0.16	0.66 ± 0.13	0.63 ± 0.12	0.46 ± 0.17	0.43 ± 0.15	0.44 ± 0.11	0.50 ± 0.06
S004	0.43 ± 0.16	0.44 ± 0.11	0.60 ± 0.19	0.54 ± 0.12	0.58 ± 0.17	0.66 ± 0.13	0.36 ± 0.15
S005	0.49 ± 0.14	0.49 ± 0.12	0.37 ± 0.17	0.60 ± 0.10	0.48 ± 0.14	0.44 ± 0.16	0.33 ± 0.16
S006	0.50 ± 0.11	0.51 ± 0.15	0.52 ± 0.09	0.47 ± 0.17	0.53 ± 0.12	0.49 ± 0.18	0.57 ± 0.18
S007	0.96 ± 0.07	0.98 ± 0.04	0.81 ± 0.10	0.91 ± 0.08	0.82 ± 0.10	0.77 ± 0.09	0.61 ± 0.17
S008	0.48 ± 0.15	0.52 ± 0.15	0.42 ± 0.18	0.38 ± 0.15	0.59 ± 0.11	0.52 ± 0.15	0.46 ± 0.13
S009	0.38 ± 0.19	0.56 ± 0.17	0.29 ± 0.11	0.37 ± 0.17	0.42 ± 0.17	0.42 ± 0.16	0.39 ± 0.16
S010	0.58 ± 0.16	0.60 ± 0.12	0.59 ± 0.16	0.58 ± 0.11	0.42 ± 0.11	0.42 ± 0.13	0.41 ± 0.07
S011	0.42 ± 0.14	0.50 ± 0.20	0.44 ± 0.12	0.41 ± 0.16	0.52 ± 0.15	0.42 ± 0.18	0.44 ± 0.19
S012	0.53 ± 0.16	0.50 ± 0.17	0.59 ± 0.17	0.50 ± 0.16	0.50 ± 0.12	0.37 ± 0.09	0.48 ± 0.17
S013	0.50 ± 0.17	0.51 ± 0.18	0.42 ± 0.14	0.49 ± 0.16	0.41 ± 0.17	0.32 ± 0.14	0.58 ± 0.17
S014	0.47 ± 0.12	0.54 ± 0.13	0.38 ± 0.11	0.53 ± 0.13	0.54 ± 0.12	0.61 ± 0.11	0.48 ± 0.14
S015	0.66 ± 0.12	0.69 ± 0.16	0.63 ± 0.14	0.44 ± 0.19	0.63 ± 0.16	0.47 ± 0.14	0.48 ± 0.13
S016	0.47 ± 0.15	0.66 ± 0.15	0.60 ± 0.11	0.58 ± 0.16	0.51 ± 0.09	0.54 ± 0.14	0.49 ± 0.16
S017	0.40 ± 0.18	0.34 ± 0.16	0.52 ± 0.15	0.36 ± 0.16	0.37 ± 0.20	0.41 ± 0.07	0.54 ± 0.17
S018	0.39 ± 0.13	0.36 ± 0.16	0.49 ± 0.11	0.51 ± 0.17	0.32 ± 0.10	0.52 ± 0.17	0.52 ± 0.15
S019	0.60 ± 0.07	0.68 ± 0.08	0.62 ± 0.07	0.68 ± 0.12	0.69 ± 0.13	0.29 ± 0.16	0.59 ± 0.12
S020	0.62 ± 0.16	0.66 ± 0.18	0.39 ± 0.15	0.61 ± 0.11	0.49 ± 0.17	0.40 ± 0.23	0.48 ± 0.10
Moyenne	0.54 ± 0.13	0.58 ± 0.14	0.53 ± 0.12	0.54 ± 0.13	0.52 ± 0.12	0.47 ± 0.11	0.49 ± 0.08

TABLE 42 – Récapitulatif — RMSE, RMS et RMS de référence ICLABEL (en μV), les 109 sujets (3 lignes par sujet), par méthode — comparable à l’ordre de grandeur des signaux (« - » = non calculé).

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S001 RMSE	24.8	24.3	17.2	16.0	19.4
S001 RMS	16.0	12.5	27.7	25.9	23.0
S001 RMS_ICLABEL	26.5	26.5	26.5	26.5	26.5
S002 RMSE	12.1	11.8	7.1	6.7	9.6
S002 RMS	9.2	7.8	11.8	12.0	12.1
S002 RMS_ICLABEL	14.7	14.7	14.7	14.7	14.7
S003 RMSE	24.5	23.6	21.5	18.9	23.7
S003 RMS	17.5	14.5	26.7	19.7	19.7
S003 RMS_ICLABEL	25.1	25.1	25.1	25.1	25.1
S004 RMSE	23.8	24.3	23.6	25.1	26.2
S004 RMS	8.7	7.3	12.2	7.5	10.5
S004 RMS_ICLABEL	26.1	26.1	26.1	26.1	26.1
S005 RMSE	14.8	14.6	12.1	12.6	14.7
S005 RMS	8.5	7.0	9.0	8.5	7.8
S005 RMS_ICLABEL	16.7	16.7	16.7	16.7	16.7
S006 RMSE	11.0	10.9	8.0	8.3	9.9
S006 RMS	6.8	6.2	7.6	6.9	8.1
S006 RMS_ICLABEL	11.8	11.8	11.8	11.8	11.8
S007 RMSE	15.6	15.0	8.2	7.9	13.2
S007 RMS	15.7	13.5	19.5	19.3	19.5
S007 RMS_ICLABEL	21.3	21.3	21.3	21.3	21.3
S008 RMSE	14.0	13.8	9.6	10.1	13.0
S008 RMS	9.5	7.8	13.0	11.2	10.5
S008 RMS_ICLABEL	16.1	16.1	16.1	16.1	16.1
S009 RMSE	25.7	25.0	19.7	19.8	25.2
S009 RMS	17.8	16.0	17.7	15.7	19.6
S009 RMS_ICLABEL	27.0	27.0	27.0	27.0	27.0
S010 RMSE	25.6	25.3	18.7	18.7	22.5
S010 RMS	16.4	12.7	25.2	24.2	22.9
S010 RMS_ICLABEL	27.6	27.6	27.6	27.6	27.6
S011 RMSE	11.5	11.5	5.7	6.7	9.5
S011 RMS	6.3	5.3	10.5	9.1	8.1
S011 RMS_ICLABEL	13.3	13.3	13.3	13.3	13.3
S012 RMSE	14.7	14.6	12.2	11.0	14.7
S012 RMS	8.4	7.0	15.4	12.5	13.4
S012 RMS_ICLABEL	15.7	15.7	15.7	15.7	15.7
S013 RMSE	27.7	27.3	20.0	20.7	25.5
S013 RMS	21.1	18.3	21.2	21.1	23.5
S013 RMS_ICLABEL	29.6	29.6	29.6	29.6	29.6
S014 RMSE	16.6	16.3	9.9	9.8	13.3
S014 RMS	13.8	11.0	16.7	17.0	16.8
S014 RMS_ICLABEL	20.7	20.7	20.7	20.7	20.7
S015 RMSE	28.5	27.6	14.6	16.0	22.8
S015 RMS	20.6	17.1	27.6	25.3	24.6
S015 RMS_ICLABEL	34.7	34.7	34.7	34.7	34.7
S016 RMSE	12.2	11.5	10.8	9.6	11.3
S016 RMS	10.1	8.1	14.0	12.2	11.2
S016 RMS_ICLABEL	12.6	12.6	12.6	12.6	12.6
S017 RMSE	34.4	33.2	27.8	28.2	33.7
S017 RMS	19.6	17.7	22.1	19.1	21.2
S017 RMS_ICLABEL	36.9	36.9	36.9	36.9	36.9
S018 RMSE	5.7	5.5	5.0	4.3	5.7
S018 RMS	3.8	3.1	6.3	5.0	5.3
S018 RMS_ICLABEL	5.8	5.8	5.8	5.8	5.8
S019 RMSE	21.8	21.6	15.8	16.7	20.2
S019 RMS	14.1	12.5	15.0	14.0	14.4
S019 RMS_ICLABEL	24.9	24.9	24.9	24.9	24.9
S020 RMSE	5.3	5.1	5.0	4.8	5.5
S020 RMS	3.6	2.9	5.8	5.3	5.1
S020 RMS_ICLABEL	5.4	5.4	5.4	5.4	5.4
S021 RMSE	11.9	11.1	9.1	8.5	10.9
S021 RMS	8.9	6.9	12.2	9.8	9.0
S021 RMS_ICLABEL	11.5	11.5	11.5	11.5	11.5
S022 RMSE	25.3	24.8	20.4	18.2	27.2
S022 RMS	16.5	14.0	27.5	21.7	25.8
S022 RMS_ICLABEL	27.7	27.7	27.7	27.7	27.7
S023 RMSE	31.7	31.1	20.0	20.6	28.3
S023 RMS	21.1	17.6	25.3	24.8	26.1
S023 RMS_ICLABEL	37.1	37.1	37.1	37.1	37.1
S024 RMSE	15.5	15.5	11.6	13.0	14.3
S024 RMS	12.4	11.9	8.6	4.5	11.0
S024 RMS_ICLABEL	14.4	14.4	14.4	14.4	14.4
S025 RMSE	7.5	7.1	4.9	4.8	6.3
S025 RMS	6.3	5.4	7.4	7.6	7.3
S025 RMS_ICLABEL	9.1	9.1	9.1	9.1	9.1
S026 RMSE	5.4	5.0	5.6	4.9	5.2
S026 RMS	4.4	3.7	6.2	5.2	4.6
S026 RMS_ICLABEL	4.9	4.9	4.9	4.9	4.9
S027 RMSE	24.5	24.0	15.0	15.9	21.2
S027 RMS	17.4	14.3	22.1	20.7	22.3
S027 RMS_ICLABEL	29.3	29.3	29.3	29.3	29.3
S028 RMSE	34.2	34.0	19.3	19.8	29.1

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S028 RMS	22.7	17.5	31.6	30.3	28.5
S028 RMS_ICLABEL	40.2	40.2	40.2	40.2	40.2
S029 RMSE	10.0	9.8	4.8	5.0	7.8
S029 RMS	8.4	7.1	11.3	11.0	10.1
S029 RMS_ICLABEL	12.8	12.8	12.8	12.8	12.8
S030 RMSE	8.1	8.1	4.7	5.1	7.1
S030 RMS	4.7	3.6	6.8	6.1	5.4
S030 RMS_ICLABEL	9.3	9.3	9.3	9.3	9.3
S031 RMSE	21.9	21.9	12.0	11.9	17.2
S031 RMS	18.4	13.6	27.3	27.9	27.8
S031 RMS_ICLABEL	27.9	27.9	27.9	27.9	27.9
S032 RMSE	30.3	30.4	20.3	23.1	28.6
S032 RMS	15.1	13.2	22.7	18.2	19.4
S032 RMS_ICLABEL	33.2	33.2	33.2	33.2	33.2
S033 RMSE	12.8	12.6	7.9	7.7	11.8
S033 RMS	8.2	7.0	12.4	11.3	12.7
S033 RMS_ICLABEL	14.8	14.8	14.8	14.8	14.8
S034 RMSE	9.2	9.2	4.6	5.0	7.0
S034 RMS	6.3	5.0	9.0	9.2	9.5
S034 RMS_ICLABEL	11.3	11.3	11.3	11.3	11.3
S035 RMSE	10.5	10.4	5.5	5.3	8.8
S035 RMS	8.2	6.6	12.1	11.9	12.1
S035 RMS_ICLABEL	13.1	13.1	13.1	13.1	13.1
S036 RMSE	27.7	26.3	21.4	20.5	25.6
S036 RMS	22.5	18.2	32.1	30.0	28.5
S036 RMS_ICLABEL	31.4	31.4	31.4	31.4	31.4
S037 RMSE	5.8	5.7	3.9	4.1	5.0
S037 RMS	3.5	3.0	4.0	3.7	3.8
S037 RMS_ICLABEL	6.6	6.6	6.6	6.6	6.6
S038 RMSE	19.1	19.1	10.5	10.5	13.9
S038 RMS	12.7	10.4	16.4	17.6	17.2
S038 RMS_ICLABEL	22.4	22.4	22.4	22.4	22.4
S039 RMSE	35.0	35.1	19.4	19.5	27.5
S039 RMS	20.5	16.0	31.2	32.5	32.0
S039 RMS_ICLABEL	40.6	40.6	40.6	40.6	40.6
S040 RMSE	18.5	18.5	14.1	15.1	17.3
S040 RMS	10.0	8.0	15.1	12.6	11.4
S040 RMS_ICLABEL	20.2	20.2	20.2	20.2	20.2
S041 RMSE	6.8	6.7	3.8	4.0	5.8
S041 RMS	4.0	3.2	6.3	6.0	6.0
S041 RMS_ICLABEL	7.9	7.9	7.9	7.9	7.9
S042 RMSE	8.2	8.0	3.4	3.2	6.0
S042 RMS	7.4	6.4	10.2	10.3	10.4
S042 RMS_ICLABEL	11.0	11.0	11.0	11.0	11.0
S043 RMSE	24.8	24.4	20.3	20.8	24.5
S043 RMS	15.4	13.0	21.8	19.5	19.8
S043 RMS_ICLABEL	26.6	26.6	26.6	26.6	26.6
S044 RMSE	33.0	31.7	23.2	21.1	31.4
S044 RMS	30.3	25.3	40.4	38.9	39.9
S044 RMS_ICLABEL	43.0	43.0	43.0	43.0	43.0
S045 RMSE	27.5	26.4	20.3	19.3	24.3
S045 RMS	18.0	14.6	24.8	23.3	23.0
S045 RMS_ICLABEL	29.0	29.0	29.0	29.0	29.0
S046 RMSE	22.8	22.3	15.4	14.5	19.0
S046 RMS	15.1	11.9	23.9	21.6	18.8
S046 RMS_ICLABEL	25.0	25.0	25.0	25.0	25.0
S047 RMSE	7.0	6.7	4.6	4.4	6.2
S047 RMS	5.6	4.4	7.6	7.2	6.2
S047 RMS_ICLABEL	8.3	8.3	8.3	8.3	8.3
S048 RMSE	42.1	40.8	21.1	20.8	32.4
S048 RMS	39.8	32.5	49.5	50.8	49.7
S048 RMS_ICLABEL	58.2	58.2	58.2	58.2	58.2
S049 RMSE	31.5	31.3	17.2	17.9	25.2
S049 RMS	16.8	14.2	27.9	26.2	26.0
S049 RMS_ICLABEL	36.0	36.0	36.0	36.0	36.0
S050 RMSE	10.1	9.5	8.1	7.7	9.1
S050 RMS	7.6	6.3	9.5	8.2	7.5
S050 RMS_ICLABEL	10.2	10.2	10.2	10.2	10.2
S051 RMSE	3.9	3.8	3.5	3.2	4.3
S051 RMS	2.2	1.9	4.4	3.6	3.9
S051 RMS_ICLABEL	4.1	4.1	4.1	4.1	4.1
S052 RMSE	30.7	30.5	17.5	17.7	24.4
S052 RMS	23.6	19.4	31.1	32.1	32.4
S052 RMS_ICLABEL	38.6	38.6	38.6	38.6	38.6
S053 RMSE	29.3	29.0	20.2	21.1	28.3
S053 RMS	18.2	15.5	24.0	20.0	24.4
S053 RMS_ICLABEL	32.9	32.9	32.9	32.9	32.9
S054 RMSE	28.0	27.2	18.0	16.9	26.7
S054 RMS	24.7	20.4	35.4	35.2	36.7
S054 RMS_ICLABEL	35.9	35.9	35.9	35.9	35.9
S055 RMSE	7.4	7.3	4.7	4.9	7.0
S055 RMS	4.1	3.5	6.9	5.7	5.2
S055 RMS_ICLABEL	8.3	8.3	8.3	8.3	8.3
S056 RMSE	33.7	32.3	23.2	23.1	29.8
S056 RMS	30.3	25.2	31.9	31.3	31.6
S056 RMS_ICLABEL	39.8	39.8	39.8	39.8	39.8

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S057 RMSE	19.4	18.6	14.3	13.7	16.2
S057 RMS	14.7	12.3	16.3	15.8	14.2
S057 RMS_ICLABEL	19.9	19.9	19.9	19.9	19.9
S058 RMSE	13.9	13.3	11.4	11.0	12.3
S058 RMS	11.2	10.2	9.8	8.4	8.9
S058 RMS_ICLABEL	12.1	12.1	12.1	12.1	12.1
S059 RMSE	22.4	21.1	21.7	21.1	21.9
S059 RMS	17.8	14.2	27.2	26.3	22.9
S059 RMS_ICLABEL	23.2	23.2	23.2	23.2	23.2
S060 RMSE	28.3	27.9	16.9	16.3	24.5
S060 RMS	18.4	15.2	24.8	25.2	25.2
S060 RMS_ICLABEL	32.9	32.9	32.9	32.9	32.9
S061 RMSE	37.2	36.4	22.4	23.7	34.3
S061 RMS	24.2	20.5	34.6	31.2	28.3
S061 RMS_ICLABEL	43.4	43.4	43.4	43.4	43.4
S062 RMSE	13.1	13.0	8.0	7.6	12.3
S062 RMS	8.8	7.3	14.7	13.7	13.3
S062 RMS_ICLABEL	15.2	15.2	15.2	15.2	15.2
S063 RMSE	12.4	11.7	10.1	9.7	11.2
S063 RMS	8.8	7.8	8.8	7.7	8.1
S063 RMS_ICLABEL	11.5	11.5	11.5	11.5	11.5
S064 RMSE	7.5	7.4	4.9	4.7	6.5
S064 RMS	4.7	3.7	8.1	7.5	6.5
S064 RMS_ICLABEL	8.6	8.6	8.6	8.6	8.6
S065 RMSE	20.9	20.2	16.1	16.3	19.1
S065 RMS	16.2	12.7	20.8	20.2	18.5
S065 RMS_ICLABEL	23.3	23.3	23.3	23.3	23.3
S066 RMSE	17.7	17.4	8.6	8.4	13.9
S066 RMS	13.4	11.0	18.7	18.3	18.7
S066 RMS_ICLABEL	21.9	21.9	21.9	21.9	21.9
S067 RMSE	6.8	6.9	5.4	5.7	6.5
S067 RMS	3.7	3.5	3.9	3.3	3.7
S067 RMS_ICLABEL	7.3	7.3	7.3	7.3	7.3
S068 RMSE	22.6	21.7	18.3	17.9	21.2
S068 RMS	15.6	12.4	21.4	19.9	19.6
S068 RMS_ICLABEL	23.2	23.2	23.2	23.2	23.2
S069 RMSE	40.7	41.8	23.8	36.6	41.3
S069 RMS	12.2	10.4	26.4	12.9	12.8
S069 RMS_ICLABEL	44.2	44.2	44.2	44.2	44.2
S070 RMSE	27.0	26.3	18.3	18.0	23.0
S070 RMS	20.5	15.9	29.6	29.6	27.9
S070 RMS_ICLABEL	31.0	31.0	31.0	31.0	31.0
S071 RMSE	18.5	17.5	14.0	13.5	15.9
S071 RMS	14.9	11.7	17.3	16.5	15.2
S071 RMS_ICLABEL	18.9	18.9	18.9	18.9	18.9
S072 RMSE	5.5	5.5	3.9	4.0	5.1
S072 RMS	3.2	2.7	5.7	5.3	5.2
S072 RMS_ICLABEL	6.1	6.1	6.1	6.1	6.1
S073 RMSE	7.1	6.7	5.4	5.2	6.3
S073 RMS	5.2	4.2	7.0	6.4	6.2
S073 RMS_ICLABEL	7.4	7.4	7.4	7.4	7.4
S074 RMSE	8.8	8.6	4.2	4.0	6.1
S074 RMS	5.9	4.8	9.0	9.1	8.7
S074 RMS_ICLABEL	10.6	10.6	10.6	10.6	10.6
S075 RMSE	44.5	44.3	23.2	24.9	36.6
S075 RMS	30.0	22.5	42.9	40.4	39.0
S075 RMS_ICLABEL	53.5	53.5	53.5	53.5	53.5
S076 RMSE	6.6	6.4	4.5	4.7	5.8
S076 RMS	4.9	4.2	6.0	5.2	4.8
S076 RMS_ICLABEL	7.5	7.5	7.5	7.5	7.5
S077 RMSE	30.7	29.9	21.7	21.1	27.3
S077 RMS	21.2	16.1	31.9	30.0	28.4
S077 RMS_ICLABEL	33.2	33.2	33.2	33.2	33.2
S078 RMSE	15.5	15.3	8.3	8.5	12.7
S078 RMS	10.8	8.6	14.9	14.0	13.5
S078 RMS_ICLABEL	18.5	18.5	18.5	18.5	18.5
S079 RMSE	35.8	35.7	20.9	25.7	30.2
S079 RMS	21.7	17.4	30.0	26.6	25.2
S079 RMS_ICLABEL	42.1	42.1	42.1	42.1	42.1
S080 RMSE	19.4	19.2	11.8	12.8	17.3
S080 RMS	11.4	9.4	16.5	14.4	14.1
S080 RMS_ICLABEL	22.2	22.2	22.2	22.2	22.2
S081 RMSE	16.3	16.3	10.1	10.1	13.6
S081 RMS	7.1	5.9	13.1	12.3	10.7
S081 RMS_ICLABEL	17.9	17.9	17.9	17.9	17.9
S082 RMSE	16.0	15.4	11.7	11.5	14.0
S082 RMS	13.4	10.8	17.7	17.6	16.5
S082 RMS_ICLABEL	18.7	18.7	18.7	18.7	18.7
S083 RMSE	28.9	28.5	17.5	18.4	25.0
S083 RMS	19.4	16.4	23.2	23.3	22.2
S083 RMS_ICLABEL	34.2	34.2	34.2	34.2	34.2
S084 RMSE	24.0	23.9	17.2	17.9	21.4
S084 RMS	16.6	13.3	19.2	20.1	20.8
S084 RMS_ICLABEL	28.8	28.8	28.8	28.8	28.8
S085 RMSE	15.5	15.5	10.3	11.1	14.2
S085 RMS	10.4	8.4	12.9	12.8	12.8

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S085 RMS_ICLABEL	18.6	18.6	18.6	18.6	18.6
S086 RMSE	39.1	38.7	27.3	29.2	34.0
S086 RMS	14.2	12.2	31.4	27.3	29.6
S086 RMS_ICLABEL	42.0	42.0	42.0	42.0	42.0
S087 RMSE	22.9	21.5	17.4	16.4	19.6
S087 RMS	19.2	15.5	23.8	23.6	22.7
S087 RMS_ICLABEL	24.9	24.9	24.9	24.9	24.9
S088 RMSE	46.9	48.2	25.8	40.2	47.0
S088 RMS	15.2	12.5	31.8	18.5	20.0
S088 RMS_ICLABEL	51.1	51.1	51.1	51.1	51.1
S089 RMSE	4.9	4.8	2.6	2.5	4.0
S089 RMS	2.9	2.3	4.8	4.5	4.4
S089 RMS_ICLABEL	5.5	5.5	5.5	5.5	5.5
S090 RMSE	20.7	19.9	19.4	18.0	22.0
S090 RMS	17.6	15.1	22.8	20.7	21.3
S090 RMS_ICLABEL	21.7	21.7	21.7	21.7	21.7
S091 RMSE	27.9	25.7	38.8	35.6	35.5
S091 RMS	21.9	18.8	41.9	37.8	33.8
S091 RMS_ICLABEL	25.4	25.4	25.4	25.4	25.4
S092 RMSE	25.8	25.0	17.6	17.5	22.1
S092 RMS	23.0	18.1	29.1	30.3	29.8
S092 RMS_ICLABEL	30.9	30.9	30.9	30.9	30.9
S093 RMSE	13.1	13.1	9.6	10.2	11.5
S093 RMS	8.2	6.6	14.8	15.0	14.0
S093 RMS_ICLABEL	14.9	14.9	14.9	14.9	14.9
S094 RMSE	30.3	30.0	18.5	18.5	26.7
S094 RMS	22.3	18.3	30.2	29.7	30.2
S094 RMS_ICLABEL	36.8	36.8	36.8	36.8	36.8
S095 RMSE	47.6	48.5	27.1	40.2	48.1
S095 RMS	24.4	19.5	34.2	21.0	16.9
S095 RMS_ICLABEL	54.5	54.5	54.5	54.5	54.5
S096 RMSE	34.7	33.8	27.0	27.8	33.7
S096 RMS	22.0	18.5	35.4	32.8	30.2
S096 RMS_ICLABEL	38.4	38.4	38.4	38.4	38.4
S097 RMSE	22.0	21.3	19.7	17.5	23.3
S097 RMS	15.4	12.7	23.7	19.4	21.4
S097 RMS_ICLABEL	22.9	22.9	22.9	22.9	22.9
S098 RMSE	8.5	8.3	4.1	3.9	6.1
S098 RMS	8.6	7.4	9.9	10.4	10.7
S098 RMS_ICLABEL	12.1	12.1	12.1	12.1	12.1
S099 RMSE	30.6	28.1	24.8	25.1	28.3
S099 RMS	26.1	21.8	22.8	20.1	21.2
S099 RMS_ICLABEL	29.9	29.9	29.9	29.9	29.9
S100 RMSE	23.8	23.2	17.6	17.6	22.3
S100 RMS	16.4	14.1	20.1	21.9	21.8
S100 RMS_ICLABEL	26.2	26.2	26.2	26.2	26.2
S101 RMSE	23.1	22.7	17.0	17.0	20.0
S101 RMS	14.1	11.3	20.3	18.5	16.7
S101 RMS_ICLABEL	24.9	24.9	24.9	24.9	24.9
S102 RMSE	6.7	6.5	4.5	4.3	5.6
S102 RMS	4.7	3.7	7.0	6.6	6.2
S102 RMS_ICLABEL	7.3	7.3	7.3	7.3	7.3
S103 RMSE	31.5	30.2	25.6	24.5	27.7
S103 RMS	26.3	20.6	37.5	36.1	32.8
S103 RMS_ICLABEL	35.7	35.7	35.7	35.7	35.7
S104 RMSE	16.9	16.7	9.0	9.3	14.9
S104 RMS	11.7	9.5	16.8	15.3	14.7
S104 RMS_ICLABEL	20.0	20.0	20.0	20.0	20.0
S105 RMSE	15.7	15.5	10.5	10.0	13.8
S105 RMS	11.7	8.9	18.8	18.2	17.7
S105 RMS_ICLABEL	18.0	18.0	18.0	18.0	18.0
S106 RMSE	25.4	25.1	13.7	14.1	20.8
S106 RMS	18.9	15.2	25.4	25.3	25.0
S106 RMS_ICLABEL	31.2	31.2	31.2	31.2	31.2
S107 RMSE	26.7	26.0	17.1	17.4	20.7
S107 RMS	20.4	16.0	25.8	24.3	22.4
S107 RMS_ICLABEL	29.2	29.2	29.2	29.2	29.2
S108 RMSE	21.4	21.2	18.7	20.1	22.4
S108 RMS	16.5	14.6	23.5	21.7	21.2
S108 RMS_ICLABEL	23.9	23.9	23.9	23.9	23.9
S109 RMSE	35.9	36.2	35.5	37.4	40.0
S109 RMS	24.0	19.1	20.4	19.5	20.1
S109 RMS_ICLABEL	44.4	44.4	44.4	44.4	44.4
Moyenne RMSE	20.8	20.4	14.4	14.8	18.8
Moyenne RMS	14.2	11.7	19.5	18.0	17.8
Moyenne RMS_ICLABEL	23.6	23.6	23.6	23.6	23.6

L'ordre de grandeur des signaux nettoyés par ICLabel est de 23.6 μ V (RMS moyen sur les 109 sujets).

TABLE 43 – Récapitulatif — RMSE, RMS et RMS de référence ICLABEL (en μV), les 109 sujets (3 lignes par sujet), par méthode — signaux de Nettoyé_Normalisé (chaque essai renormalisé avec la moyenne/écart-type du signal brut Prétraité_Total du même sujet, « - » = non calculé).

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S001 RMSE	39.5	39.4	22.8	22.0	27.7
S001 RMS	36.2	36.2	36.2	36.2	36.2
S001 RMS_ICLABEL	36.2	36.2	36.2	36.2	36.2
S002 RMSE	17.5	16.7	9.3	8.8	13.0
S002 RMS	18.7	18.7	18.7	18.7	18.7
S002 RMS_ICLABEL	18.7	18.7	18.7	18.7	18.7
S003 RMSE	60.5	60.1	44.8	43.4	56.2
S003 RMS	54.1	54.1	54.1	54.1	54.1
S003 RMS_ICLABEL	54.1	54.1	54.1	54.1	54.1
S004 RMSE	35.3	36.6	34.8	39.1	41.7
S004 RMS	32.7	32.7	32.7	32.7	32.7
S004 RMS_ICLABEL	32.7	32.7	32.7	32.7	32.7
S005 RMSE	23.3	22.7	16.7	17.9	22.9
S005 RMS	22.4	22.4	22.4	22.4	22.4
S005 RMS_ICLABEL	22.4	22.4	22.4	22.4	22.4
S006 RMSE	22.6	22.6	14.9	15.5	19.5
S006 RMS	20.6	20.6	20.6	20.6	20.6
S006 RMS_ICLABEL	20.6	20.6	20.6	20.6	20.6
S007 RMSE	23.1	22.1	11.6	11.2	18.7
S007 RMS	28.7	28.7	28.7	28.7	28.7
S007 RMS_ICLABEL	28.7	28.7	28.7	28.7	28.7
S008 RMSE	25.6	25.0	16.1	16.8	23.4
S008 RMS	25.3	25.3	25.3	25.3	25.3
S008 RMS_ICLABEL	25.3	25.3	25.3	25.3	25.3
S009 RMSE	62.3	61.4	45.9	45.6	60.6
S009 RMS	56.7	56.7	56.7	56.7	56.7
S009 RMS_ICLABEL	56.7	56.7	56.7	56.7	56.7
S010 RMSE	43.1	43.6	27.8	28.4	35.3
S010 RMS	39.7	39.7	39.7	39.7	39.7
S010 RMS_ICLABEL	39.7	39.7	39.7	39.7	39.7
S011 RMSE	16.0	15.9	6.9	7.7	12.5
S011 RMS	15.8	15.8	15.8	15.8	15.8
S011 RMS_ICLABEL	15.8	15.8	15.8	15.8	15.8
S012 RMSE	40.9	40.9	29.2	27.9	37.4
S012 RMS	36.5	36.5	36.5	36.5	36.5
S012 RMS_ICLABEL	36.5	36.5	36.5	36.5	36.5
S013 RMSE	60.0	60.5	41.3	43.6	53.8
S013 RMS	56.3	56.3	56.3	56.3	56.3
S013 RMS_ICLABEL	56.3	56.3	56.3	56.3	56.3
S014 RMSE	22.9	22.4	12.3	12.4	17.3
S014 RMS	25.4	25.4	25.4	25.4	25.4
S014 RMS_ICLABEL	25.4	25.4	25.4	25.4	25.4
S015 RMSE	39.4	37.4	17.3	18.3	29.9
S015 RMS	42.3	42.3	42.3	42.3	42.3
S015 RMS_ICLABEL	42.3	42.3	42.3	42.3	42.3
S016 RMSE	26.4	25.9	20.1	19.4	23.9
S016 RMS	24.8	24.8	24.8	24.8	24.8
S016 RMS_ICLABEL	24.8	24.8	24.8	24.8	24.8
S017 RMSE	70.1	67.2	52.3	52.2	68.0
S017 RMS	63.3	63.3	63.3	63.3	63.3
S017 RMS_ICLABEL	63.3	63.3	63.3	63.3	63.3
S018 RMSE	15.8	15.5	11.5	10.8	14.2
S018 RMS	13.8	13.8	13.8	13.8	13.8
S018 RMS_ICLABEL	13.8	13.8	13.8	13.8	13.8
S019 RMSE	38.1	37.7	24.3	26.0	34.5
S019 RMS	37.4	37.4	37.4	37.4	37.4
S019 RMS_ICLABEL	37.4	37.4	37.4	37.4	37.4
S020 RMSE	12.7	12.7	9.9	10.1	11.8
S020 RMS	11.1	11.1	11.1	11.1	11.1
S020 RMS_ICLABEL	11.1	11.1	11.1	11.1	11.1
S021 RMSE	26.7	26.4	18.0	18.2	24.3
S021 RMS	23.2	23.2	23.2	23.2	23.2
S021 RMS_ICLABEL	23.2	23.2	23.2	23.2	23.2
S022 RMSE	73.6	72.2	52.7	50.3	71.4
S022 RMS	68.5	68.5	68.5	68.5	68.5
S022 RMS_ICLABEL	68.5	68.5	68.5	68.5	68.5
S023 RMSE	55.9	54.2	30.5	31.2	48.9
S023 RMS	56.2	56.2	56.2	56.2	56.2
S023 RMS_ICLABEL	56.2	56.2	56.2	56.2	56.2
S024 RMSE	39.8	40.5	31.3	36.5	38.2
S024 RMS	34.6	34.6	34.6	34.6	34.6
S024 RMS_ICLABEL	34.6	34.6	34.6	34.6	34.6
S025 RMSE	11.1	10.4	6.7	6.6	8.9
S025 RMS	11.8	11.8	11.8	11.8	11.8
S025 RMS_ICLABEL	11.8	11.8	11.8	11.8	11.8
S026 RMSE	13.8	13.4	11.7	11.6	13.0
S026 RMS	11.9	11.9	11.9	11.9	11.9
S026 RMS_ICLABEL	11.9	11.9	11.9	11.9	11.9
S027 RMSE	40.3	39.0	22.2	23.6	34.5
S027 RMS	41.7	41.7	41.7	41.7	41.7

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S027 RMS_ICLABEL	41.7	41.7	41.7	41.7	41.7
S028 RMSE	54.5	53.3	27.3	27.9	45.1
S028 RMS	55.1	55.1	55.1	55.1	55.1
S028 RMS_ICLABEL	55.1	55.1	55.1	55.1	55.1
S029 RMSE	12.7	12.4	5.5	5.6	9.3
S029 RMS	14.5	14.5	14.5	14.5	14.5
S029 RMS_ICLABEL	14.5	14.5	14.5	14.5	14.5
S030 RMSE	11.0	10.9	5.3	5.5	8.8
S030 RMS	10.9	10.9	10.9	10.9	10.9
S030 RMS_ICLABEL	10.9	10.9	10.9	10.9	10.9
S031 RMSE	28.8	28.2	14.4	14.2	19.9
S031 RMS	33.1	33.1	33.1	33.1	33.1
S031 RMS_ICLABEL	33.1	33.1	33.1	33.1	33.1
S032 RMSE	58.6	59.1	34.3	38.2	53.5
S032 RMS	53.9	53.9	53.9	53.9	53.9
S032 RMS_ICLABEL	53.9	53.9	53.9	53.9	53.9
S033 RMSE	25.8	25.3	14.1	13.7	22.2
S033 RMS	25.8	25.8	25.8	25.8	25.8
S033 RMS_ICLABEL	25.8	25.8	25.8	25.8	25.8
S034 RMSE	11.3	11.2	4.8	5.3	7.8
S034 RMS	12.4	12.4	12.4	12.4	12.4
S034 RMS_ICLABEL	12.4	12.4	12.4	12.4	12.4
S035 RMSE	17.1	16.7	8.3	8.0	13.3
S035 RMS	18.8	18.8	18.8	18.8	18.8
S035 RMS_ICLABEL	18.8	18.8	18.8	18.8	18.8
S036 RMSE	53.0	50.9	36.4	35.8	45.9
S036 RMS	53.4	53.4	53.4	53.4	53.4
S036 RMS_ICLABEL	53.4	53.4	53.4	53.4	53.4
S037 RMSE	8.4	8.2	4.6	5.0	6.9
S037 RMS	8.3	8.3	8.3	8.3	8.3
S037 RMS_ICLABEL	8.3	8.3	8.3	8.3	8.3
S038 RMSE	26.1	26.0	12.1	12.8	17.9
S038 RMS	26.6	26.6	26.6	26.6	26.6
S038 RMS_ICLABEL	26.6	26.6	26.6	26.6	26.6
S039 RMSE	51.3	50.7	24.5	25.2	37.2
S039 RMS	51.4	51.4	51.4	51.4	51.4
S039 RMS_ICLABEL	51.4	51.4	51.4	51.4	51.4
S040 RMSE	28.7	28.8	19.7	21.5	26.0
S040 RMS	26.4	26.4	26.4	26.4	26.4
S040 RMS_ICLABEL	26.4	26.4	26.4	26.4	26.4
S041 RMSE	10.8	10.5	5.4	5.6	8.8
S041 RMS	10.9	10.9	10.9	10.9	10.9
S041 RMS_ICLABEL	10.9	10.9	10.9	10.9	10.9
S042 RMSE	10.5	10.1	3.9	3.8	6.9
S042 RMS	12.6	12.6	12.6	12.6	12.6
S042 RMS_ICLABEL	12.6	12.6	12.6	12.6	12.6
S043 RMSE	52.7	52.4	39.1	41.6	49.4
S043 RMS	48.0	48.0	48.0	48.0	48.0
S043 RMS_ICLABEL	48.0	48.0	48.0	48.0	48.0
S044 RMSE	59.1	56.3	39.4	36.5	53.3
S044 RMS	67.9	67.9	67.9	67.9	67.9
S044 RMS_ICLABEL	67.9	67.9	67.9	67.9	67.9
S045 RMSE	51.4	50.2	34.3	33.0	42.3
S045 RMS	46.4	46.4	46.4	46.4	46.4
S045 RMS_ICLABEL	46.4	46.4	46.4	46.4	46.4
S046 RMSE	38.5	38.1	22.9	22.1	30.4
S046 RMS	36.4	36.4	36.4	36.4	36.4
S046 RMS_ICLABEL	36.4	36.4	36.4	36.4	36.4
S047 RMSE	11.5	11.1	6.9	6.7	9.9
S047 RMS	12.0	12.0	12.0	12.0	12.0
S047 RMS_ICLABEL	12.0	12.0	12.0	12.0	12.0
S048 RMSE	49.8	46.1	22.4	22.4	36.1
S048 RMS	63.3	63.3	63.3	63.3	63.3
S048 RMS_ICLABEL	63.3	63.3	63.3	63.3	63.3
S049 RMSE	48.7	47.5	23.1	23.6	36.9
S049 RMS	47.9	47.9	47.9	47.9	47.9
S049 RMS_ICLABEL	47.9	47.9	47.9	47.9	47.9
S050 RMSE	19.2	18.7	14.2	14.2	17.3
S050 RMS	17.3	17.3	17.3	17.3	17.3
S050 RMS_ICLABEL	17.3	17.3	17.3	17.3	17.3
S051 RMSE	12.3	12.2	9.0	9.2	12.1
S051 RMS	10.9	10.9	10.9	10.9	10.9
S051 RMS_ICLABEL	10.9	10.9	10.9	10.9	10.9
S052 RMSE	41.0	40.1	21.1	21.6	30.7
S052 RMS	45.7	45.7	45.7	45.7	45.7
S052 RMS_ICLABEL	45.7	45.7	45.7	45.7	45.7
S053 RMSE	65.4	64.9	40.7	41.2	60.3
S053 RMS	62.9	62.9	62.9	62.9	62.9
S053 RMS_ICLABEL	62.9	62.9	62.9	62.9	62.9
S054 RMSE	50.6	48.8	30.7	29.0	44.5
S054 RMS	57.8	57.8	57.8	57.8	57.8
S054 RMS_ICLABEL	57.8	57.8	57.8	57.8	57.8
S055 RMSE	15.3	15.1	8.6	8.8	14.2
S055 RMS	14.6	14.6	14.6	14.6	14.6
S055 RMS_ICLABEL	14.6	14.6	14.6	14.6	14.6
S056 RMSE	57.0	55.6	37.4	37.1	49.6

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S056 RMS	61.0	61.0	61.0	61.0	61.0
S056 RMS_ICLABEL	61.0	61.0	61.0	61.0	61.0
S057 RMSE	36.2	36.0	24.8	24.3	30.0
S057 RMS	33.1	33.1	33.1	33.1	33.1
S057 RMS_ICLABEL	33.1	33.1	33.1	33.1	33.1
S058 RMSE	30.7	30.5	26.3	26.6	29.7
S058 RMS	25.6	25.6	25.6	25.6	25.6
S058 RMS_ICLABEL	25.6	25.6	25.6	25.6	25.6
S059 RMSE	41.8	41.0	33.0	33.0	37.1
S059 RMS	39.1	39.1	39.1	39.1	39.1
S059 RMS_ICLABEL	39.1	39.1	39.1	39.1	39.1
S060 RMSE	48.9	47.7	25.8	24.7	40.5
S060 RMS	49.3	49.3	49.3	49.3	49.3
S060 RMS_ICLABEL	49.3	49.3	49.3	49.3	49.3
S061 RMSE	64.9	62.5	35.3	35.9	57.8
S061 RMS	65.7	65.7	65.7	65.7	65.7
S061 RMS_ICLABEL	65.7	65.7	65.7	65.7	65.7
S062 RMSE	27.7	27.5	15.1	14.6	24.7
S062 RMS	28.0	28.0	28.0	28.0	28.0
S062 RMS_ICLABEL	28.0	28.0	28.0	28.0	28.0
S063 RMSE	28.1	27.5	22.4	22.0	25.6
S063 RMS	23.3	23.3	23.3	23.3	23.3
S063 RMS_ICLABEL	23.3	23.3	23.3	23.3	23.3
S064 RMSE	13.6	13.4	8.0	7.9	11.5
S064 RMS	13.4	13.4	13.4	13.4	13.4
S064 RMS_ICLABEL	13.4	13.4	13.4	13.4	13.4
S065 RMSE	33.1	32.7	23.3	23.9	29.0
S065 RMS	32.6	32.6	32.6	32.6	32.6
S065 RMS_ICLABEL	32.6	32.6	32.6	32.6	32.6
S066 RMSE	26.8	25.9	11.7	11.6	20.3
S066 RMS	28.9	28.9	28.9	28.9	28.9
S066 RMS_ICLABEL	28.9	28.9	28.9	28.9	28.9
S067 RMSE	12.8	12.9	8.9	9.5	11.9
S067 RMS	11.5	11.5	11.5	11.5	11.5
S067 RMS_ICLABEL	11.5	11.5	11.5	11.5	11.5
S068 RMSE	44.4	44.2	32.4	32.8	39.3
S068 RMS	39.6	39.6	39.6	39.6	39.6
S068 RMS_ICLABEL	39.6	39.6	39.6	39.6	39.6
S069 RMSE	56.8	60.3	24.4	40.7	59.4
S069 RMS	52.8	52.8	52.8	52.8	52.8
S069 RMS_ICLABEL	52.8	52.8	52.8	52.8	52.8
S070 RMSE	43.2	42.6	26.3	25.8	33.8
S070 RMS	43.6	43.6	43.6	43.6	43.6
S070 RMS_ICLABEL	43.6	43.6	43.6	43.6	43.6
S071 RMSE	31.0	30.9	22.1	22.0	26.4
S071 RMS	28.7	28.7	28.7	28.7	28.7
S071 RMS_ICLABEL	28.7	28.7	28.7	28.7	28.7
S072 RMSE	11.2	11.1	7.0	7.3	9.8
S072 RMS	10.5	10.5	10.5	10.5	10.5
S072 RMS_ICLABEL	10.5	10.5	10.5	10.5	10.5
S073 RMSE	13.7	13.5	9.3	9.5	11.7
S073 RMS	12.6	12.6	12.6	12.6	12.6
S073 RMS_ICLABEL	12.6	12.6	12.6	12.6	12.6
S074 RMSE	10.6	10.0	4.5	4.3	6.7
S074 RMS	11.3	11.3	11.3	11.3	11.3
S074 RMS_ICLABEL	11.3	11.3	11.3	11.3	11.3
S075 RMSE	65.1	63.0	29.8	31.3	51.9
S075 RMS	68.1	68.1	68.1	68.1	68.1
S075 RMS_ICLABEL	68.1	68.1	68.1	68.1	68.1
S076 RMSE	10.6	10.2	6.6	6.8	9.2
S076 RMS	10.5	10.5	10.5	10.5	10.5
S076 RMS_ICLABEL	10.5	10.5	10.5	10.5	10.5
S077 RMSE	59.4	59.2	37.0	36.8	49.0
S077 RMS	55.8	55.8	55.8	55.8	55.8
S077 RMS_ICLABEL	55.8	55.8	55.8	55.8	55.8
S078 RMSE	24.7	24.0	11.9	11.9	19.8
S078 RMS	25.6	25.6	25.6	25.6	25.6
S078 RMS_ICLABEL	25.6	25.6	25.6	25.6	25.6
S079 RMSE	43.2	42.1	20.7	26.2	33.3
S079 RMS	44.1	44.1	44.1	44.1	44.1
S079 RMS_ICLABEL	44.1	44.1	44.1	44.1	44.1
S080 RMSE	32.7	32.1	17.2	18.2	28.1
S080 RMS	32.0	32.0	32.0	32.0	32.0
S080 RMS_ICLABEL	32.0	32.0	32.0	32.0	32.0
S081 RMSE	24.6	24.3	13.1	12.8	19.0
S081 RMS	22.8	22.8	22.8	22.8	22.8
S081 RMS_ICLABEL	22.8	22.8	22.8	22.8	22.8
S082 RMSE	23.9	23.2	16.0	15.7	19.6
S082 RMS	25.0	25.0	25.0	25.0	25.0
S082 RMS_ICLABEL	25.0	25.0	25.0	25.0	25.0
S083 RMSE	44.0	43.1	21.8	23.6	36.6
S083 RMS	45.6	45.6	45.6	45.6	45.6
S083 RMS_ICLABEL	45.6	45.6	45.6	45.6	45.6
S084 RMSE	31.3	30.7	19.9	21.4	26.5
S084 RMS	33.0	33.0	33.0	33.0	33.0
S084 RMS_ICLABEL	33.0	33.0	33.0	33.0	33.0

Sujet	DuoCL	GCTNet	ICUNet	ICUNet++	ICUNet_attn
S085 RMSE	23.6	23.3	13.8	15.3	21.2
S085 RMS	24.8	24.8	24.8	24.8	24.8
S085 RMS_ICLABEL	24.8	24.8	24.8	24.8	24.8
S086 RMSE	67.3	65.0	40.3	43.3	52.7
S086 RMS	59.4	59.4	59.4	59.4	59.4
S086 RMS_ICLABEL	59.4	59.4	59.4	59.4	59.4
S087 RMSE	37.1	35.9	25.7	24.5	29.8
S087 RMS	36.4	36.4	36.4	36.4	36.4
S087 RMS_ICLABEL	36.4	36.4	36.4	36.4	36.4
S088 RMSE	67.4	71.3	30.7	40.1	68.8
S088 RMS	64.4	64.4	64.4	64.4	64.4
S088 RMS_ICLABEL	64.4	64.4	64.4	64.4	64.4
S089 RMSE	8.8	8.6	4.2	3.9	7.0
S089 RMS	8.6	8.6	8.6	8.6	8.6
S089 RMS_ICLABEL	8.6	8.6	8.6	8.6	8.6
S090 RMSE	58.2	58.0	48.6	47.8	57.5
S090 RMS	55.3	55.3	55.3	55.3	55.3
S090 RMS_ICLABEL	55.3	55.3	55.3	55.3	55.3
S091 RMSE	98.9	95.9	90.9	91.0	99.3
S091 RMS	84.1	84.1	84.1	84.1	84.1
S091 RMS_ICLABEL	84.1	84.1	84.1	84.1	84.1
S092 RMSE	37.7	37.2	23.9	23.5	29.5
S092 RMS	40.3	40.3	40.3	40.3	40.3
S092 RMS_ICLABEL	40.3	40.3	40.3	40.3	40.3
S093 RMSE	22.5	22.8	15.6	16.8	19.2
S093 RMS	21.8	21.8	21.8	21.8	21.8
S093 RMS_ICLABEL	21.8	21.8	21.8	21.8	21.8
S094 RMSE	48.4	47.5	26.3	26.2	41.1
S094 RMS	52.4	52.4	52.4	52.4	52.4
S094 RMS_ICLABEL	52.4	52.4	52.4	52.4	52.4
S095 RMSE	58.6	59.5	23.9	36.9	57.0
S095 RMS	57.8	57.8	57.8	57.8	57.8
S095 RMS_ICLABEL	57.8	57.8	57.8	57.8	57.8
S096 RMSE	72.4	70.9	49.5	51.6	65.8
S096 RMS	69.0	69.0	69.0	69.0	69.0
S096 RMS_ICLABEL	69.0	69.0	69.0	69.0	69.0
S097 RMSE	61.1	60.4	47.0	45.6	58.7
S097 RMS	55.3	55.3	55.3	55.3	55.3
S097 RMS_ICLABEL	55.3	55.3	55.3	55.3	55.3
S098 RMSE	9.8	9.2	3.9	4.0	6.4
S098 RMS	12.8	12.8	12.8	12.8	12.8
S098 RMS_ICLABEL	12.8	12.8	12.8	12.8	12.8
S099 RMSE	59.9	57.8	50.2	52.5	59.2
S099 RMS	54.6	54.6	54.6	54.6	54.6
S099 RMS_ICLABEL	54.6	54.6	54.6	54.6	54.6
S100 RMSE	48.0	47.4	32.7	32.7	42.3
S100 RMS	45.9	45.9	45.9	45.9	45.9
S100 RMS_ICLABEL	45.9	45.9	45.9	45.9	45.9
S101 RMSE	35.4	35.1	23.4	23.5	28.9
S101 RMS	32.3	32.3	32.3	32.3	32.3
S101 RMS_ICLABEL	32.3	32.3	32.3	32.3	32.3
S102 RMSE	11.5	11.4	6.9	6.7	8.9
S102 RMS	10.9	10.9	10.9	10.9	10.9
S102 RMS_ICLABEL	10.9	10.9	10.9	10.9	10.9
S103 RMSE	48.1	47.1	34.1	33.3	39.2
S103 RMS	48.8	48.8	48.8	48.8	48.8
S103 RMS_ICLABEL	48.8	48.8	48.8	48.8	48.8
S104 RMSE	30.0	29.2	14.2	14.1	25.7
S104 RMS	30.9	30.9	30.9	30.9	30.9
S104 RMS_ICLABEL	30.9	30.9	30.9	30.9	30.9
S105 RMSE	30.2	30.1	18.0	17.5	24.1
S105 RMS	30.1	30.1	30.1	30.1	30.1
S105 RMS_ICLABEL	30.1	30.1	30.1	30.1	30.1
S106 RMSE	38.8	37.9	18.7	19.4	30.6
S106 RMS	41.8	41.8	41.8	41.8	41.8
S106 RMS_ICLABEL	41.8	41.8	41.8	41.8	41.8
S107 RMSE	40.1	40.1	24.1	24.5	30.3
S107 RMS	38.7	38.7	38.7	38.7	38.7
S107 RMS_ICLABEL	38.7	38.7	38.7	38.7	38.7
S108 RMSE	52.8	53.0	40.7	44.4	52.2
S108 RMS	51.9	51.9	51.9	51.9	51.9
S108 RMS_ICLABEL	51.9	51.9	51.9	51.9	51.9
S109 RMSE	42.3	41.5	40.6	44.1	49.3
S109 RMS	46.7	46.7	46.7	46.7	46.7
S109 RMS_ICLABEL	46.7	46.7	46.7	46.7	46.7
Moyenne RMSE	36.5	36.0	23.1	23.8	31.9
Moyenne RMS	35.8	35.8	35.8	35.8	35.8
Moyenne RMS_ICLABEL	35.8	35.8	35.8	35.8	35.8

L'ordre de grandeur des signaux (Nettoyé_Normalisé) est de 35.8 μ V (RMS moyen sur les 109 sujets).